

Inflation Risk and Yield Spread Changes

Diego Bonelli*

Abstract

Inflation risk explains a significant share of the systematic residual variation in yield spread changes beyond standard credit factors and intermediation frictions. Expected inflation, inflation uncertainty, and inflation cyclicalities help account for bond-level movements and the residual common component in corporate credit. The effect is stronger for firms closer to default, such as highly leveraged and lower-rated issuers, and for firms with thinner margins or greater price pass-through, and weaker when debt adjusts more quickly through refinancing or floating-rate borrowing. Together, these patterns link residual spread commonality to heterogeneous exposure to inflation risk.

JEL classification: G10; G12; G20

Keywords: Inflation Risk, Corporate Bonds, Yield Spread Changes, Inflation-Linked Derivatives.

*Banco de España, Alcalá 48, 28007 Madrid, Spain. *Email:* diego.bonelli@bde.es. The views expressed in this manuscript are those of the author and do not necessarily represent the views of Banco de España or the Eurosystem. I thank Ravi Bansal, Andrea Eisfeldt, Nils Friewald, Ljubica Georgievska, Lars Lochstoer, Francis Longstaff, Denis Mokanov, Jens Dick-Nielsen, Galo Nuño, Giorgio Ottonello, Svein-Arne Persson, Walter Pohl, Hiroatsu Tanaka, Jinyuan Zhang, several conference audiences, and seminar participants at the Norwegian School of Economics, UCLA Anderson, Banco de España, Rotterdam School of Management, and Universidad de Navarra for helpful discussions and comments. Part of the work on this paper was completed while visiting UCLA Anderson. First draft: December 2022. This draft: September 2026

I. Introduction

Corporate yield spread changes remain empirically challenging to explain. [Collin-Dufresne, Goldstein and Martin \(2001\)](#) (hereafter [CDGM](#)) show that fundamental credit-risk variables play a significant role, yet a substantial amount of unexplained variation persists. Much of this unexplained variation loads on a common component, pointing to aggregate factors alongside standard credit-risk variables. Given the predominance of nominal debt in the U.S. corporate sector and the slow adjustment of firms’ leverage, changes in the inflation outlook should affect corporate credit risk by changing the real value of promised payments.

In this paper, I investigate whether inflation risk helps explain the large residual systematic variation in yield spread changes. I measure inflation risk using three inflation-related variables, primarily constructed from inflation swap rates: expected inflation, inflation uncertainty, and inflation cyclicalities. In yield spread change regressions, these proxies account for more than a quarter of the unexplained variation beyond standard credit-risk and intermediation factors and are significantly related to the common residual component.

These variables are motivated by models of nominal debt and default. In [Bhamra, Fisher and Kuehn \(2011\)](#), [Gomes, Jermann and Schmid \(2016\)](#), [Gomes and Schmid \(2021\)](#), and [Bhamra, Dorion, Jeanneret and Weber \(2022\)](#), inflation affects corporate credit risk because liabilities are fixed in nominal terms while leverage adjusts slowly. Expected inflation therefore changes the real value of promised debt payments and, through this channel, default probabilities and credit spreads. Inflation uncertainty captures a different mechanism. In [David \(2007\)](#) and [Kang and Pflueger \(2015\)](#), defaultable debt contains an option-like component, where greater uncertainty about nominal firm value increases the value of the default option and widens spreads.¹ The credit-risk implications of inflation also depend on the macroeconomic state. Inflation that accompanies stronger expected real activity has differ-

¹A more indirect channel is suggested by [Fischer \(2016\)](#). In that setting, long-run inflation uncertainty can lower bond values by distorting investment decisions, as price uncertainty delays investment or reallocates it toward less productive projects. As in [Baldwin and Ruback \(1986\)](#), higher uncertainty raises the relative value of short-lived assets, all else equal, which can increase default risk and widen yield spreads.

ent implications from inflation, or disinflation, that accompanies weak real activity. The two imply different effects on real debt burdens and on repayment capacity, as emphasized by [Campbell, Pflueger and Viceira \(2020\)](#), [Fang, Liu and Roussanov \(2025\)](#), and [Bonelli, Palazzo and Yamarthy \(2025\)](#).² These theories guide the choice of empirical proxies and the cross-sectional analysis. Inflation risk should have stronger effects for firms whose balance sheets, debt contracts, and operating cash flows leave their credit risk more exposed to nominal conditions.

I construct the inflation risk measures from zero-coupon inflation swaps.³ Since swaps are quoted at multiple maturities, I use the inflation swap curve to match each bond’s duration to the corresponding inflation horizon. Expected inflation is measured as the monthly change in the duration-matched swap rate. Inflation uncertainty is the change in the rolling 21-day volatility of that rate, while inflation cyclicalities is the change in the rolling correlation between expected-inflation changes and revisions in expected real GDP growth from Consensus Economics.

I document three main empirical results. First, the inflation-risk proxies are significantly related to bond-level yield spread changes in a sample of U.S. corporate bonds from Academic TRACE covering September 2004 to December 2021. Expected inflation is negatively related to spread changes, consistent with higher inflation reducing the real burden of outstanding nominal debt. Inflation uncertainty is positively related to spread changes, consistent with a default-option channel in which greater uncertainty about nominal firm value widens spreads. Inflation cyclicalities is positive but weaker, consistent with a channel whose effect depends on the macroeconomic state. Taken together, the three proxies explain more than one-third of the residual variation left by standard credit-risk variables. They continue to explain more than one-quarter after adding the intermediation controls of [Friewald and Nagler \(2019\)](#), [He,](#)

²[Chen \(2010\)](#) and [Bhamra, Dorion, Jeanneret and Weber \(2022\)](#) also note that recovery rates tend to be lower in low-inflation economic states.

³Inflation swaps are among the most liquid over-the-counter inflation-linked derivatives and are widely used to measure inflation compensation across maturities (e.g., [Fleming and Sporn \(2013\)](#); [Fleckenstein, Longstaff and Lustig \(2016\)](#); [Diercks, Campbell, Sharpe and Soques \(2023\)](#)).

Khorrami and Song (2022), and Eisfeldt, Herskovic and Liu (2024) (henceforth FN, HKS, and EHL). The economic magnitudes are also meaningful, with a one-standard-deviation increase in expected inflation narrowing yield spreads by about 8 basis points and a one-standard-deviation increase in inflation uncertainty widening them by about 5 basis points.

Second, inflation risk helps explain the common component in residual spread changes. Following CDGM, I regress monthly yield spread changes on variables motivated by structural credit-risk models. In recent U.S. corporate bond data, these variables explain an important but limited share of monthly spread changes, with a mean adjusted R^2 of 35.5%. The remaining variation is highly systematic, with the first principal component explaining 79.7% of systematic residual variation after aggregating residuals into maturity-leverage cohorts. The three inflation-risk proxies jointly explain about 21% of the variation in the first principal component of the CDGM residuals. This relation remains economically meaningful after controlling for intermediation, macroeconomic, monetary-policy, and growth-expectation variables.

Third, I document substantial heterogeneity across firms, consistent with inflation risk operating through nominal liabilities, contract flexibility, and operating exposure. If expected inflation affects credit risk by changing the real value of nominal liabilities, its effect should be stronger when default risk is more sensitive to the debt burden. Consistent with this mechanism, spreads are more sensitive to inflation risk for firms with high leverage, weak profitability, low profit margins, and lower credit quality. The same mechanism also implies weaker sensitivity when nominal liabilities adjust more quickly. Consistent with this prediction, firms with greater refinancing intensity or more floating-rate debt are less sensitive to expected inflation.⁴ The cross-sectional evidence also points to a cash-flow channel alongside the liability channel. Inflation affects credit risk not only through the real value of debt, but also through input costs, output prices, and operating cash flows. Firms with low markups

⁴The floating-rate debt and refinancing variables proxy for how quickly nominal debt adjusts. The mechanism does not require direct indexation to inflation; firms with faster liability repricing are less exposed to changes in the real value of fixed nominal debt.

are more sensitive to expected inflation, while firms with greater price pass-through are more sensitive to both expected inflation and inflation uncertainty. I also decompose the residual variation explained by the inflation-risk proxies to identify which dimension of inflation risk contributes most across these groups. Expected inflation is the largest contributor in every sort, with particularly large differences across groups in the leverage and credit-rating sorts, where its contribution rises sharply for the most levered and lowest-rated issuers. Among the operating sorts, expected inflation remains dominant, while inflation uncertainty plays a larger role for firms with greater price pass-through.

The evidence holds across measurement choices and specifications. Results are similar when using survey-anchored expectations, constructed by fitting a daily Nelson-Siegel-Svensson curve following [Reis \(2020\)](#) to swaps, TIPS breakevens, and survey data. The expected-inflation and inflation-uncertainty signs survive a wide range of alternative measures and controls: fixed-maturity swaps, TIPS breakevens, CPI-based proxies, realized inflation, forecast disagreement, firm-level equity volatility, and duration-matched real yields. The findings remain similar under alternative residual groupings, panel specifications, clustering schemes, and an extended sample through 2024 using Enhanced TRACE. The time-series evidence also shows that sensitivity to expected inflation strengthens when inflation risk is salient, sensitivity to inflation uncertainty rises when the outlook is harder to forecast, and inflation cyclicalities are weaker on average and varies across macroeconomic states.

The evidence therefore shows that inflation risk explains part of the variation in yield spread changes left by standard credit-risk and intermediation variables. This explanatory power is not uniform across firms. Expected inflation matters most when nominal liabilities are large or slow to adjust, while operating characteristics such as markup and price pass-through shape which dimension of inflation risk matters. These results link residual spread commonality to inflation risk and show that its effects vary with nominal debt exposure, debt-contract flexibility, and firms' operating exposure to inflation.

This paper contributes to three related strands of the literature. First, it contributes to

the literature on inflation and corporate credit. A large literature studies inflation as an aggregate risk factor in equity and bond markets, including [Fama \(1981\)](#), [Chen, Roll and Ross \(1986\)](#), [Weber \(2014\)](#), [Eraker, Shaliastovich and Wang \(2015\)](#), [Fleckenstein, Longstaff and Lustig \(2017\)](#), and [Boons, Duarte, de Roon and Szymanowska \(2020\)](#). In corporate credit, existing work shows that inflation affects default risk and bond prices through nominal liabilities, inflation uncertainty, and the relation between inflation and real activity, as in [David \(2007\)](#), [Kang and Pflueger \(2015\)](#), [Bhamra, Fisher and Kuehn \(2011\)](#), [Gomes, Jerermann and Schmid \(2016\)](#), [Gomes and Schmid \(2021\)](#), and [Bhamra, Dorion, Jeanneret and Weber \(2022\)](#). More recent work studies related channels in corporate bond returns, including [Ceballos \(2021\)](#) and [Lu, Nozawa and Song \(2025\)](#). I add to this literature by showing that inflation risk helps explain variation in individual corporate bond yield spread changes, including the residual variation left by standard credit-risk variables and intermediation variables.

Second, the paper contributes to the literature on the common component in corporate bond spread changes. [CDGM](#) show that standard credit-risk variables leave a large common residual component in yield spread changes. Other work attributes this residual commonality to liquidity, market structure, and intermediary constraints, including [Friewald and Nagler \(2019\)](#), [He, Khorrami and Song \(2022\)](#), and [Eisfeldt, Herskovic and Liu \(2024\)](#). I show that inflation risk explains part of this common residual component, linking residual spread commonality to aggregate nominal risk as well as to market frictions and intermediation.

Third, the paper contributes to the literature on the transmission of inflation risk across firms. The nominal-debt literature shows that inflation affects default risk by changing the real value of promised debt payments, as in [Bhamra, Fisher and Kuehn \(2011\)](#), [Gomes, Jerermann and Schmid \(2016\)](#), [Gomes and Schmid \(2021\)](#), and [Bhamra, Dorion, Jeanneret and Weber \(2022\)](#), while theories of debt maturity and refinancing emphasize how contract structure shapes firms' sensitivity to shocks, including [Diamond \(1991\)](#), [Goldstein, Ju and Leland \(2001\)](#), and [Friewald, Nagler and Wagner \(2022\)](#). I bring these mechanisms to bond-level

yield spread changes and show that the effects are stronger for firms whose liability structure and operating exposure leave credit risk more sensitive to nominal conditions. The variance decomposition further shows that the relative importance of expected inflation and inflation uncertainty differs across balance-sheet, debt-contract, and operating characteristics.

The remainder of the paper is organized as follows. Section II outlines the data sources. Section III introduces the inflation-risk proxies, presents the main results, and reports the survey-anchored specifications. Section IV discusses the heterogeneity results. Section V provides additional evidence and robustness tests. Section VI concludes.

II. Data

The sample of industrial corporate bond transactions comes from the Academic Trade Reporting and Compliance Engine (TRACE), maintained by the Financial Industry Regulatory Authority (FINRA). The main sample covers September 2004 to December 2021. In the robustness analysis, I extend the sample through December 2024 using the Enhanced TRACE database.⁵ I follow the cleaning steps from Dick-Nielsen and Poulsen (2019), thus cleaning same-day corrections and cancellations, removing reversals, as well as double counting of agency trades. Then, I apply a median filter and a reversal filter to eliminate further potential data errors following Edwards, Harris and Piwowar (2007).⁶ I merge corporate bond pricing data with the Mergent Fixed Income Securities Database (FISD) to obtain bond characteristics, such as offering amount, offering date, maturity, coupon rate, bond rating, bond option features, and issuer information, and with CRSP/Compustat to obtain firm characteristics.⁷

⁵The main sample ends in December 2021 because the Enhanced TRACE database lacks dealer identifiers and therefore does not permit the construction of the full set of OTC market proxies used in the baseline specifications. In the extended sample, I use only the subset of OTC variables that can be constructed without dealer identifiers.

⁶The median filter eliminates any transaction where the price deviates by more than 10% from the daily median or from a nine-trading-day median centered at the trading day. The reversal filter eliminates any transaction with an absolute price change that deviates from the lead, lag, and average lead/lag price change by at least 10%.

⁷See the Internet Appendix for the construction of the TRACE/CRSP linking table. The matching uses

Following the literature on corporate bonds, I restrict the sample to corporate debentures and exclude bonds with variable coupons, convertibility, putability, asset-backed status, exchangeability, private placements, perpetual terms, preferred securities, secured lease obligations, missing ratings, or foreign-currency denomination. I also remove bonds issued by financial firms (SIC 6000–6999) and utility firms (SIC 4900–4999), and bonds with issue sizes under \$10 million or a time to maturity of more than 30 years or less than one month.⁸

The main variable in the empirical analysis is the monthly change in the corporate bond yield spread. Using TRACE intraday data, I first eliminate transactions with when-issued, lock-in, special trades, or primary trades flags. I then calculate the daily clean price as the volume-weighted average of intraday prices to reduce the effect of bid-ask bounce and individual trade noise, following Bessembinder, Kahle, Maxwell and Xu (2008). For each month, I use the observation closest to the last trading day of the month within a five-trading-day window. I compute the end-of-month corporate bond yield from the volume-weighted price and define the yield spread as the difference between this yield and the yield of a risk-free bond with the same cash-flow structure as the corporate bond. The risk-free benchmark is derived from the U.S. Treasury yield curve estimates published by the Federal Reserve Board.⁹ I compute monthly changes and returns of all variables. To avoid asynchronicity issues, I match the dates of variables available at the daily frequency, such as the VIX and Treasury rates, to the dates on which end-of-month bond prices are measured. I retain only bonds with at least 25 observations of monthly yield spread changes.

Following CDGM, I construct market and firm-specific variables that, according to structural models, determine yield spread changes. These include the Standard & Poor’s 500 index return (RM_t) from CRSP, changes in the VIX index (ΔVIX_t) from the Chicago Board Options Exchange, and changes in Treasury rates and the term-structure slope (ΔRF_t , ΔRF_t^2 ,

issuer CUSIPs, parent CUSIPs, trading symbols, and firm names, and accounts for mergers, delistings, and ticker changes.

⁸The expected-inflation and inflation-uncertainty results remain robust when all bonds, irrespective of industry or bond type, are included, while inflation cyclicalities loses significance.

⁹Yield spreads are winsorized at the 0.5% level.

and $\Delta Slope_t$) from the off-the-run yield curves of [Gürkaynak, Sack and Wright \(2007\)](#). As a systematic proxy for the probability or magnitude of a downward jump in firm value ($\Delta Jump_t$), I construct a measure based on at- and out-of-the-money put options and at- and in-the-money call options with maturities of less than one year, traded on the SPX index. The option data come from OptionMetrics and the Options Price Reporting Authority (OPRA). For the exact procedure used to estimate the jump component, I refer to [CDGM](#).

I use market leverage as a proxy for firm creditworthiness. Following [Friewald and Nagler \(2019\)](#), market leverage ($\Delta Lev_{i,t}$) is book debt divided by the sum of book debt and the market value of equity. Book debt is Compustat long-term debt (DLTT) plus debt in current liabilities (DLC). I lag debt-related accounting information by six months to avoid using data before it is plausibly available to investors. Market equity is common shares outstanding times the CRSP share price.

The inflation-related variables are constructed from inflation-linked market prices and survey expectations. I obtain daily bid and ask quotes for U.S. zero-coupon inflation swaps from Bloomberg for maturities between one and thirty years starting in July 2004.¹⁰ Inflation swaps are forward contracts in which the inflation buyer pays a fixed nominal rate and receives a payment linked to realized inflation. They provide market-based inflation compensation across maturities and are the main source for the baseline inflation measures.

I also use zero-coupon TIPS yields and breakeven inflation rates from [Gürkaynak, Sack and Wright \(2010\)](#), which are derived from TIPS coupon bond yields. Since both inflation swaps and TIPS are indexed to the non-seasonally adjusted CPI-U, I seasonally adjust both curves following [Fleckenstein, Longstaff and Lustig \(2014\)](#).¹¹ For each corporate bond, I match the bond's duration to the corresponding point on the inflation curve, using spline

¹⁰I use the liquid benchmark maturities from one to ten years, and the 12-, 15-, 20-, and 30-year maturities. Bloomberg does not retain inflation swap quotes before July 2004. I use mid quotes, computed as the average of bid and ask quotes.

¹¹Seasonal adjustment starts at the shortest available maturity, one year for zero-coupon inflation swaps and two years for TIPS breakevens. This matters because duration matching may evaluate the inflation curve at fractional maturities, where unadjusted rates can mechanically reflect seasonal CPI patterns. The full procedure is described in the Internet Appendix.

interpolation when the required maturity is not directly quoted. Section III defines the three baseline inflation-risk proxies constructed from this duration-matched curve.

I also construct survey-anchored inflation-expectation measures. I use the Michigan Survey of Consumers, the Survey of Professional Forecasters, and Consensus Economics inflation forecasts. Following [Dovern, Fritsche and Slacalek \(2012\)](#), I convert current-year and next-year Consensus Economics forecasts into fixed 12-month-ahead expectations by weighting the two calendar-year forecasts according to the share of the forecast horizon falling in each year.¹² I combine inflation swaps, TIPS breakevens, and survey expectations in a Nelson-Siegel-Svensson curve fitted daily to the inflation term structure. Survey observations are aligned to their publication dates and carried forward between releases using exponential time-decay weights. Evaluating the fitted curve at each bond’s duration yields a bond-specific duration-matched expected-inflation series. This provides a complementary measure of inflation expectations that does not rely on any single market or survey series. For the cyclical measure, I also use Consensus Economics forecasts of real GDP growth.

Lastly, Producer Price Index (PPI) data used to construct industry-level price pass-through come from the U.S. Bureau of Labor Statistics, while the input-output tables come from the U.S. Bureau of Economic Analysis. Data on realized inflation, unemployment, real consumption, and income come from the Federal Reserve Bank of St. Louis. I construct intermediation proxies using intraday Academic TRACE data following [FN](#) and [EHL](#). To construct the distress factor of [HKS](#), I use the intermediary capital ratio from Zhiguo He’s website and the noise measure from Jun Pan’s website. For the exact procedures, I refer to [FN](#), [HKS](#), and [EHL](#).

Table I Panel A reports the summary statistics of the sample of corporate bonds. The sample consists of 449788 observations of monthly yield spread changes for 6826 bonds issued by 936 firms. The average yield spread is 2.32%, with a standard deviation of 3.05%. The average offering size is \$742 million, and the average time to maturity is approximately nine

Table I
about
here

¹²This procedure avoids discontinuities that arise from using calendar-year forecasts directly.

years. Around 24% of the observations are high-yield bonds.

III. Inflation Risk and Yield Spread Changes

This section examines whether inflation risk helps explain the residual component in corporate yield spread changes. The starting point is the benchmark evidence in [CDGM](#). Standard credit-risk variables explain an important part of spread movements, but leave a large residual component that is highly systematic across bonds. I use this residual component as the object of interest and ask whether it is related to expected inflation, inflation uncertainty, and inflation cyclicalities.

I first introduce the inflation-risk proxies and the channels they are meant to capture. I then estimate their relation to yield spread changes after controlling for the standard credit-risk variables of [CDGM](#) and the intermediation factors of [FN](#), [HKS](#), and [EHL](#). Next, I examine whether the results are sensitive to the measurement of inflation expectations using survey-anchored term-structure estimates. Finally, I assess the robustness of the findings under alternative estimation and inference procedures.

A. Inflation-Risk Proxies

Inflation expectations should be measured at the horizon that matters for each bond. A given inflation-horizon shock should not affect short- and long-duration bonds equally. I therefore construct the main baseline expected-inflation and uncertainty proxies from duration-matched inflation swap rates. Inflation swaps provide forward-looking, market-based inflation compensation across maturities, which allows me to match each bond's duration to the corresponding point on the inflation curve. I complement these bond-level measures with a time-series proxy for inflation cyclicalities, constructed from the relation between long-term expected inflation and expected real GDP growth. Together, these variables capture three inflation-related dimensions: expected inflation, inflation uncertainty, and the

macroeconomic state in which inflation news arrives.

Inflation swaps are well suited for the baseline analysis because they are observed at high frequency and provide inflation rates across maturities. I therefore use them as the main source of forward-looking inflation variation. Since traded inflation-linked instruments may also contain risk-premium and liquidity components, Section III.C reports a complementary analysis using survey-anchored Nelson-Siegel-Svensson fitted expectations.

Expected inflation. The first proxy captures innovations in expected inflation. The channel follows from the nominal nature of corporate debt. Corporate bonds promise fixed nominal payments, while firms adjust leverage only gradually. Higher expected inflation reduces the real burden of outstanding debt, lowers real leverage, and should compress yield spreads. This prediction follows from models with nominal debt and sticky leverage, including Bhamra, Fisher and Kuehn (2011), Gomes, Jermann and Schmid (2016), Gomes and Schmid (2021), and Bhamra, Dorion, Jeanneret and Weber (2022). Conversely, declines in expected inflation raise the real burden of debt and can strengthen debt-overhang forces. In Gomes, Jermann and Schmid (2016), this mechanism distorts investment and production decisions and raises credit spreads.

Expected inflation can also coincide with changes in expected cash flows. Over much of the sample, expected inflation is positively related to expected real activity, so increases in expected inflation often occur together with stronger expected fundamentals (David and Veronesi (2013); Campbell, Pflueger and Viceira (2020); Bonelli, Palazzo and Yamarthy (2025)).¹³

I measure expected-inflation innovations with the monthly change in the duration-matched inflation swap rate, denoted by $\Delta E[\mu^S]_{i,t}$. For each bond, I evaluate the zero-coupon inflation swap curve at the bond’s duration and use the resulting rate as the relevant inflation horizon. This procedure matches short-duration bonds to short-horizon inflation compensation and

¹³Bonelli, Palazzo and Yamarthy (2025) show that the sign of inflation sensitivity depends on the relation between inflation and expected growth. In the full sample, expected inflation is, on average, positively related to expected growth, so the unconditional expected-inflation coefficient is predicted to be negative.

long-duration bonds to longer-horizon measures. The expected coefficient on $(\Delta E[\mu^S]_{i,t})$ is negative.¹⁴

Inflation uncertainty. The second proxy captures uncertainty about the inflation outlook. Its effect differs from the expected-inflation channel. While higher expected inflation lowers the real value of nominal debt, greater uncertainty about future inflation increases the dispersion of nominal firm value. In [David \(2007\)](#) and [Kang and Pflueger \(2015\)](#), defaultable debt can be viewed as a risk-free bond minus a put option on firm value. Higher uncertainty raises the value of this default option and widens spreads.

I measure this channel with the monthly change in the rolling standard deviation of the duration-matched inflation swap rate, denoted by $\Delta\sigma_{i,t}^S$. For each bond-month, I compute the 21-trading-day rolling standard deviation of the daily matched swap rate, evaluated at the bond’s month-end pricing date, and then take the monthly change. The measure is intended to capture short-horizon variation in the market-implied inflation path over the trailing month. Section [V.B](#) examines whether this variation instead reflects within-month drift in expected inflation. The expected coefficient on $\Delta\sigma_{i,t}^S$ is positive.¹⁵

Inflation cyclical. The third proxy captures the macroeconomic state in which inflation news arrives. Inflation has different credit-risk implications depending on whether it is associated with stronger or weaker expected real activity. Inflation news that arrives with stronger expected real activity differs from inflation, or disinflation, that arrives in weak real states because it implies different effects on real debt burdens and repayment capacity. This state dependence is emphasized by [Kang and Pflueger \(2015\)](#), [Bhamra, Dorion, Jeanneret](#)

¹⁴Because measured leverage enters directly in the benchmark regressions, the coefficient on expected inflation captures variation in spreads that is not summarized by contemporaneous leverage alone. Expected-inflation innovations can affect the real value of outstanding liabilities, expected nominal cash flows, refinancing conditions, and default risk before these effects are fully reflected in measured leverage. This distinction is important because the empirical tests ask whether the inflation-risk proxies add explanatory power beyond the standard [CDGM](#) controls, including leverage.

¹⁵This proxy is distinct from forecast disagreement. Disagreement measures dispersion in beliefs across forecasters, while $\Delta\sigma_{i,t}^S$ measures time-series variation in the market-implied inflation outlook. I use disagreement-based measures as part of the robustness analysis.

and Weber (2022), Campbell, Pflueger and Viceira (2020), Fang, Liu and Roussanov (2025), and Bonelli, Palazzo and Yamarchy (2025).

I capture this channel with an expectations-based cyclical proxy, denoted by $\Delta\rho_t^S$. The level (ρ_t^S) is the rolling 24-month correlation between changes in long-horizon expected inflation and revisions in expected real GDP growth.¹⁶ Expected growth revisions are measured using the Consensus Economics fixed-horizon 12-month-ahead GDP forecast.

The proxy ρ_t^S captures the hedge value of inflation for nominal corporate debt. A higher correlation between expected inflation and expected growth means that weak growth states are more likely to coincide with lower inflation and, therefore, with a higher real value of nominal debt. An increase in ρ_t^S , and hence in $\Delta\rho_t^S$, should therefore be associated with wider spreads. Unlike expected inflation and inflation uncertainty, this channel is inherently state-dependent. A pooled coefficient may average across regimes in which inflation has different credit-risk implications, so I interpret the cyclical estimates together with the time-series evidence in Section V.E.

Table I Panel B reports the unconditional correlations between yield spread changes and the three inflation-risk proxies, as well as the correlations among the proxies.

B. Baseline Results

I begin by documenting the residual variation in recent U.S. corporate bond data. Following CDGM, I estimate a separate time-series regression for each bond,

$$\Delta YS_{i,t} = \alpha_i + \boldsymbol{\beta}_i^T \boldsymbol{\Delta S}_{i,t} + \varepsilon_{i,t},$$

where $\Delta YS_{i,t}$ is the monthly yield spread change and $\boldsymbol{\Delta S}_{i,t}$ contains the standard structural credit-risk controls. These controls include changes in leverage, Treasury rates, squared

¹⁶For the cyclical proxy, expected inflation is measured using the 10-year inflation swap rate. I use the 10-year zero-coupon inflation swap rate when available and the 10-year TIPS breakeven rate otherwise. The results are similar when expected inflation is instead measured using the Nelson-Siegel-Svensson fitted inflation curve described in Section III.C.

changes in Treasury rates, changes in the slope of the term structure, changes in the VIX, the S&P 500 return, and the jump-risk proxy.

Column (1) of Table II reports the benchmark results.¹⁷ The mean adjusted R^2 is 35.5%. Thus, standard credit-risk variables explain an important part of monthly spread movements, but close to two-thirds of the variation remains unexplained. This magnitude is in line with the evidence in CDGM and with recent work.

Table II
about
here

The residual variation is also highly systematic. To measure this commonality, I assign bonds to 18 maturity-leverage cohorts. The maturity groups are below five years, five to eight years, and above eight years; the leverage groups are below 15%, 15%–25%, 25%–35%, 35%–45%, 45%–55%, and above 55%. For each cohort and month, I compute the average residual $\epsilon_{g,t}$ and extract principal components from the covariance matrix of the cohort-level residuals.¹⁸

The principal component analysis confirms that this residual variation is highly concentrated in one common component. The total unexplained variance, measured by the trace of the cohort residual covariance matrix, is about 112 in basis points. The first principal component accounts for 79.7% of systematic residual variation, while the second accounts for only 5.4%. The residuals are therefore not simply idiosyncratic bond-level noise. They contain a common component that remains after standard credit-risk variables have been removed. I use this common component to test whether inflation risk accounts for part of the systematic variation left unexplained by the benchmark credit-risk specification.

I add the inflation-risk proxies to the benchmark specification:

$$\Delta YS_{i,t} = \alpha_i + \beta_i^T \Delta \mathbf{S}_{i,t} + \theta_i^T \Delta \mathbf{I}_{i,t} + \Gamma_i^T \Delta \mathbf{C}_{i,t} + v_{i,t},$$

where $\Delta \mathbf{I}_{i,t}$ contains $\Delta E[\mu^S]_{i,t}$, $\Delta \sigma_{i,t}^S$, and $\Delta \rho_t^S$. The vector $\Delta \mathbf{C}_{i,t}$ contains the OTC market

¹⁷Detailed coefficient estimates for the CDGM replication and the corresponding cohort-level PCA loadings are reported in Internet Appendix.

¹⁸The PCA is performed on cohort-month average residuals rather than on individual bond residuals, following CDGM. This reduces the influence of idiosyncratic bond-level noise. Section V shows that the results are similar under alternative cohort definitions.

friction controls from [FN](#), the intermediary distress factors from [HKS](#), and interdealer price dispersion from [EHL](#).

Panel A of Table [II](#) reports average bond-level coefficients, t-statistics, and mean and median adjusted R^2 values. Following [CDGM](#), inference is based on the cross-sectional dispersion of the bond-level estimates. I compute the t-statistic as the average coefficient divided by the standard deviation of the bond-level coefficients, scaled by the square root of the number of bonds.¹⁹ Columns (2) to (4) add the inflation-risk proxies one at a time, column (5) includes them jointly, and columns (6) to (8) add the intermediation controls.

The signs are stable across specifications. Increases in expected inflation are associated with narrower spreads, and increases in inflation uncertainty with wider spreads. The cyclicity proxy also enters positively, but its average effect is weak and, as Section [III.D](#) and Section [V.E](#) show, it is significant in some samples and inference procedures but not others. This is consistent with the proxy capturing regime-dependent variation that a single full-sample coefficient cannot summarize well.

I use three residual measures in the analysis. The first is the bond-level residual from the spread-change regressions. The second is the cohort-level residual, which averages these residuals within the 18 maturity-leverage cohorts. This is the object used to compute the fraction of residual variation explained. The third is the common residual component, measured as the first principal component of the cohort-level residuals. In each case, explanatory power is measured relative to the corresponding specification without the inflation-risk proxies. When intermediation controls are included, the comparison is therefore to the residual variance left after adding those controls, but before adding expected inflation, inflation uncertainty, and inflation cyclicity.

The inflation-risk proxies account for a sizeable share of the residual variation. Following [HKS](#), I measure explanatory power using the fraction of variation explained, defined as the

¹⁹This procedure is standard in this literature, including [CDGM](#), [FN](#), [HKS](#), and [EHL](#). Since the sample contains 6826 bonds, the resulting t-statistics can be large in absolute value. Section [III.D](#) verifies the main results using panel fixed-effects regressions with alternative clustering schemes.

reduction in residual variance after adding the inflation-risk proxies. For each cohort g , let $\varepsilon_{g,t}$ denote the residual from the specification without inflation-risk proxies and $v_{g,t}$ the residual from the corresponding specification that includes them. The fraction of variation explained is

$$\text{FVE} = 1 - \frac{\sum_{g=1}^{18} \sigma_{v_g}^2}{\sum_{g=1}^{18} \sigma_{\varepsilon_g}^2}, \quad (1)$$

where $\sigma_{v_g}^2$ and $\sigma_{\varepsilon_g}^2$ are the time-series variances of the residuals for cohort g in the specifications with and without inflation-risk proxies, respectively.

The three inflation-risk proxies jointly explain 36.6% of residual variation before adding intermediation controls. Their explanatory power remains economically large after controlling for OTC frictions, intermediary distress, and interdealer price dispersion. In the fully controlled specification in column (8), they explain 27.0% of the remaining residual variation. Thus, inflation risk explains additional residual spread variation beyond the intermediation factors emphasized in recent work.

Panel C of Table II turns to the common component of the residuals. I regress the first principal component of the CDGM residuals on the inflation-risk proxies.²⁰ The inflation-risk proxies are jointly significant and explain 21.2% of the variation in PC1 based on adjusted R^2 before adding intermediation controls, and 5.8% in the fully controlled specification. Inflation risk therefore helps explain not only bond-level residual variation but also the common residual component.

The economic magnitudes are also substantial. I use the full specification in column (8) and compute the implied yield spread change from a one-standard-deviation change in each inflation proxy. A one-standard-deviation increase in expected inflation, $\Delta E[\mu^S]_{i,t}$, narrows spreads by 8.2 basis points. A one-standard-deviation increase in inflation uncertainty, $\Delta \sigma_{i,t}^S$, widens spreads by about 5.2 basis points, while the effect of inflation cyclicalities, $\Delta \rho_t^S$, is smaller, at about 0.6 basis points. These effects are sizeable. The magnitude for expected

²⁰Because PC1 is a monthly time-series factor, I use monthly cross-sectional averages of expected inflation and inflation uncertainty. The results are similar when fixed-maturity inflation swap rates are used instead.

inflation is close to that of a one-standard-deviation change in the 10-year Treasury rate, which is around 9 basis points. It is also large relative to average spreads, as a one-standard-deviation increase in the swap rate reduces the mean yield spread by 3.5% and the median yield spread by 5.7%.

C. Survey-Anchored Inflation Expectations

The baseline analysis uses duration-matched inflation swap rates because swaps provide high-frequency, forward-looking inflation compensation across maturities. This term-structure information is useful for matching inflation exposure to each bond’s duration. At the same time, swap rates are traded prices and may embed risk premia and liquidity components in addition to expected inflation. I therefore construct a complementary measure that preserves the maturity information in market prices while anchoring the curve with survey expectations.

Following the approach in [Reis \(2020\)](#), I combine market-based inflation compensation with survey expectations in a daily fitted inflation curve. Specifically, I fit a Nelson-Siegel-Svensson curve, following [Nelson and Siegel \(1987\)](#) and [Svensson \(1994\)](#), to inflation swap rates, TIPS breakevens, and inflation forecasts from the Michigan Survey of Consumers, the Survey of Professional Forecasters, and Consensus Economics. Because the surveys are released on different dates and at different frequencies, I align each observation to its publication date and carry it forward between releases using exponential time-decay weights. More recent survey readings therefore receive more weight in the daily curve.²¹

Evaluating the fitted curve at each bond’s duration gives a daily, bond-specific measure of expected inflation. From this fitted series, I construct the same inflation-risk proxies used in the swap-based analysis. The first, $\Delta E[\mu^E]_{i,t}$, is the monthly change in fitted expected inflation. The second, $\Delta \sigma_{i,t}^E$, is the change in the rolling 21-day volatility of the fitted series. The third, $\Delta \rho_t^E$, is the change in the rolling 24-month correlation between fitted expected-

²¹In the baseline implementation, the half-life is 30 days for the Survey of Professional Forecasters and 15 days for the Michigan Survey of Consumers and Consensus Economics.

inflation changes and revisions in the Consensus Economics GDP forecast.

Table III reports the results. Every column includes the CDGM structural controls and the intermediation controls, so column (6), which adds the three survey-anchored proxies jointly, is the counterpart of column (8) of Table II. The signs are unchanged. Expected inflation remains negatively related to yield spread changes, while inflation uncertainty remains positively related. The cyclical proxy also enters positively.

Table
III
about
here

The explanatory power is close to the swap-based benchmark. Jointly, the survey-anchored proxies explain 24.3% of residual variation, compared with 27.0% in the baseline swap specification. The expected-inflation estimate is somewhat smaller and the uncertainty estimate about half the baseline, consistent with survey-anchored expectations being smoother and containing less high-frequency variation than market prices, while the cyclical estimate is larger and more statistically significant. The relation between inflation risk and yield spread changes therefore survives when the market-based term structure is anchored with survey information.

D. Estimation Approach and Inference

In the baseline estimates in Tables II and III, I estimate a separate time-series regression for each bond and then average the coefficients across bonds. This approach allows each bond to have its own sensitivity to the credit-risk controls and inflation-risk proxies, which is useful in a market with substantial heterogeneity across issuers and bond characteristics. Statistical significance is computed from the cross-sectional dispersion of the bond-level estimates, following CDGM. As EHL note, however, the bond-level time-series betas can be noisy, which may affect the standard errors of the average coefficients. Since the sample also contains several thousand bonds, the resulting t-statistics can be large in absolute value.

Table IV examines whether the main conclusions depend on this estimation approach. Column (1) repeats the mean-group estimates from column (8) of Table II. Columns (2) to (4) pool all bond-month observations in panel fixed-effects regressions with bond fixed

Table
IV
about
here

effects. These specifications impose common slope coefficients across bonds and differ only in the clustering of standard errors. Column (2) double-clusters by bond and month, column (3) clusters by month, and column (4) clusters by bond.²²

The point estimates are stable across the two approaches. The expected-inflation coefficient is -0.326 under the mean-group estimator and -0.339 under panel fixed effects. The corresponding coefficients on inflation uncertainty are 0.714 and 0.587 . Expected inflation and inflation uncertainty remain statistically significant under double clustering. The cyclicity proxy remains positive but is no longer statistically significant under double clustering, as expected of a state-dependent effect; Section V.E documents that state dependence directly.

IV. Heterogeneity of Inflation Risk

The baseline results show that inflation risk explains a sizeable share of residual yield spread variation. I next examine whether this relation varies systematically across firms and bonds. The analysis distinguishes between two related but different questions. The first is whether yield spreads are more sensitive to inflation risk for firms with particular balance-sheet, operating, or debt-contract characteristics. The second is whether these differences in sensitivity translate into differences in the amount of residual spread variation explained by inflation risk.

The cross-sectional tests examine firm and bond characteristics that shape exposure to inflation risk through balance sheets, operating activity, and debt structure. I focus on measures of financial strength, operating characteristics, liability adjustment, and bond characteristics. For each characteristic, I estimate a bond-fixed-effects panel regression in which the coefficients on the three inflation-risk proxies are allowed to vary across groups, while the coefficients on the structural credit-risk variables and intermediation controls remain

²²The panel fixed-effects estimator is less flexible than the mean-group estimator because it imposes homogeneous slopes across bonds. I therefore use it as an inference check rather than as the main specification.

common across groups:

$$\Delta YS_{i,t} = \alpha_i + \beta^T \Delta S_{i,t} + \Gamma^T \Delta C_{i,t} + \sum_g (\theta_g^T \Delta I_{i,t}) \mathbb{1}\{G_{i,t} = g\} + v_{i,t},$$

where $\Delta S_{i,t}$ contains the structural credit-risk variables of [CDGM](#), $\Delta C_{i,t}$ contains the intermediation controls used in the baseline specification, and $\Delta I_{i,t}$ contains $\Delta E[\mu^S]_{i,t}$, $\Delta \sigma_{i,t}^S$, and $\Delta \rho_i^S$. Groups are formed using information available before the yield spread observation.²³ I use a one-month lag for the time-varying characteristics, except for price pass-through and debt growth, which are based on expanding averages through twelve months before the observation.

Table [V](#) reports the High-Minus-Low difference, $\theta_{g_{\max}} - \theta_{g_{\min}}$, for each characteristic, together with its two-way clustered t-statistics. These estimates provide a direct test of whether inflation-risk sensitivities differ between the extreme groups. The patterns in Panel A of Table [V](#) are strongest for profitability and credit quality. Expected-inflation sensitivity is more negative for firms with low return on assets, low profit margins, and lower credit ratings, while inflation uncertainty is more positive for the same groups. These differences are statistically significant and suggest that inflation risk matters more when repayment capacity is weaker and default risk is more exposed to the real burden of nominal debt. Smaller firms are also more sensitive to expected inflation, although size is not associated with significant differences in inflation uncertainty or cyclicalities. Inflation cyclicalities varies with some measures of financial strength, but the pattern is less systematic.

Table [V](#)
about
here

Panel B shows that operating characteristics provide an additional source of heterogeneity. Low-markup firms have a more negative expected-inflation coefficient than high-markup

²³Return on assets is net income over total assets, profit margin is net income over sales, markup is $(Sales - COGS)/Sales$, and operating leverage is $(SG\&A + Depreciation)/(SG\&A + Depreciation + COGS)$. Re-financing intensity is $(DD1 + DD2 + DD3)/(DD1 + DLTT)$, and debt growth is the quarterly change in $(DLTT + DLC)/AT$. Size is the offering amount at issuance, leverage is defined in Section [II](#), and credit rating and maturity come from Mergent FISD. Floating-rate exposure identifies issuers with at least 5% of debt at floating rates, and trading volume is monthly TRACE volume. Price pass-through is constructed at the industry level from PPI data and input-output linkages as the absolute ratio of own-industry output-price inflation to input-weighted supplier-price inflation, averaged over the expanding window used to form the groups.

firms, while differences in inflation uncertainty and cyclicalities are much smaller. Because firms with thin markups price close to marginal cost, they pass cost changes into output prices more fully, so inflation raises their nominal revenues more and has larger effects on operating cash flows and, in turn, credit risk. Price pass-through captures a related but distinct dimension of operating exposure.²⁴ Firms in the highest pass-through group have a more negative expected-inflation coefficient and a more positive inflation-uncertainty coefficient than firms in the lowest group, although the pattern is not monotone across intermediate groups. This indicates greater sensitivity to both expected inflation and inflation uncertainty when output prices move more relative to input costs.²⁵ Operating leverage, by contrast, shows little systematic heterogeneity.

The debt-structure results in Panel C provide more direct evidence on the real debt-burden channel. Highly levered firms have a more negative expected-inflation coefficient and a larger positive inflation-uncertainty coefficient, with significant High-Minus-Low differences for all three inflation-risk proxies, although only at the 10% level for inflation uncertainty. Together with the profitability and credit-rating results, this pattern suggests that inflation-risk sensitivity is stronger when credit risk depends more heavily on the real burden of debt.

The remaining debt-structure measures show how this exposure changes when liabilities adjust more quickly. Firms that refinance more frequently are less sensitive to expected inflation and inflation uncertainty, consistent with debt terms being reset more often. Firms with greater floating-rate exposure are also less sensitive to both proxies. Although the bonds in the sample have fixed coupons, some issuers rely partly on floating-rate debt elsewhere in their capital structure, so this pattern is consistent with faster repricing weakening the real debt-burden channel. Recent debt growth points in the opposite direction: firms with

²⁴Because it requires a match to industry-level producer-price and input-output data, this sort covers 108686 bond-month observations on 2749 bonds, roughly a quarter of the panel.

²⁵In the Internet Appendix I show that markup and price pass-through capture related but separate dimensions of operating exposure. The markup effect is present within both low- and high-pass-through groups, while the pass-through effect is most pronounced among low-markup firms, with the greatest expected-inflation sensitivity occurring when low markups and high pass-through coincide. The pass-through effect is broadly similar across credit-quality groups and within the manufacturing sector. Moreover, gross margins rise with producer-price inflation in high-pass-through industries but remain flat elsewhere.

greater debt growth are more sensitive to expected inflation, while differences in inflation uncertainty and cyclicalities are not statistically significant.

The final two rows of Panel C consider maturity and trading volume. Maturity shows no clear monotone pattern. Expected-inflation and inflation-uncertainty sensitivities are strongest at intermediate maturities, but the extreme groups do not differ significantly. This may partly reflect the duration matching of the inflation-risk proxies, which already aligns the relevant inflation horizon with each bond’s maturity exposure. Inflation cyclicalities are the exception, with significantly lower sensitivity for long- than for short-maturity bonds. Trading volume provides a separate check on whether the expected-inflation result is driven by illiquidity. Sensitivity is stronger among more actively traded bonds, while inflation uncertainty and cyclicalities do not differ significantly across the extreme groups. The stronger relation among actively traded bonds therefore weighs against an illiquidity-based explanation.

I next examine whether differences in sensitivity translate into differences in the amount of residual variation explained by inflation risk, and which inflation-risk component drives those differences. For each sorting characteristic, I add the three inflation-risk proxies separately to the control specification and measure their contribution by the reduction in group-level residual variance.²⁶

Figure 1
about
here

Figure 1 shows that expected inflation is the largest contributor across all characteristics, while inflation cyclicalities generally play a smaller role. The expected-inflation contribution varies most across price pass-through, leverage, credit rating, profitability, and profit margin, and is notably smaller for the highest-rated issuers. These patterns reinforce the interpretation that the expected-inflation channel is most important when firms are more exposed through their liabilities or operating conditions.

²⁶I form portfolios of control-model residuals by group and month and measure the reduction in each group’s residual variance from adding a given proxy, expressed relative to the summed residual variance across groups within the same sorting characteristic. Single-proxy contributions need not sum to the joint contribution because the proxies are correlated. In Figure 1, stacked bars report the single-proxy contributions and diamonds the joint contribution.

The decomposition also clarifies that stronger sensitivity does not necessarily imply greater explanatory power. Some characteristics generate sizeable coefficient differences but only modest changes in the fraction of residual variation explained, particularly among the operating characteristics. The coefficient estimates therefore identify where inflation-risk exposure differs, while the decomposition shows where those differences matter quantitatively. Taken together, the evidence points to both liability and operating channels, with expected inflation accounting for most of the cross-sectional variation in explanatory power.

V. Additional Evidence and Robustness

Having shown that inflation-risk proxies explain residual movements in corporate bond spreads, I next assess the robustness of the results to alternative measures, controls, residual constructions, and sample periods. I consider a broad set of inflation-risk proxies, compare their explanatory power with that of standard macroeconomic and monetary-policy variables, and revisit the construction of the residual common component under alternative cohort definitions. I also extend the analysis through 2024 using Enhanced TRACE. Across these exercises, the signs and significance of expected inflation and inflation uncertainty are stable, although magnitudes vary with the measure used, while inflation cyclicalities are weaker on average and vary more across economic conditions.

A. *Alternative Measures of Inflation Risk*

The baseline analysis uses duration-matched zero-coupon inflation swaps to construct the inflation-risk proxies. This choice matches each corporate bond to the relevant point on the inflation curve and preserves the maturity-specific nature of the exposure. Section III.C has already shown that the results are similar when the inflation curve is anchored with survey expectations. This subsection asks whether the evidence survives other ways of measuring inflation risk: fixed-maturity swaps, TIPS breakevens, CPI-based proxies, and alternative

constructions of the uncertainty and cyclical proxy.

Table VI reports the results. Column (1) includes the full set of CDGM credit-risk controls and the intermediation controls. Column (2) repeats the baseline specification using duration-matched inflation swaps. Column (3) uses the 10-year inflation swap rate for all bonds, removing bond-specific duration matching while staying within the same inflation-swap market. Column (4) uses seasonally adjusted TIPS breakeven rates from Gürkaynak, Sack and Wright (2010), matched to bond duration. Column (5) uses CPI-based proxies constructed from realized inflation, following Boons, Duarte, de Roon and Szymanowska (2020). Because the CPI-based measures are built from realized inflation rather than from inflation-linked security prices, they are free of any risk-premium or liquidity component in the swap market.

Table VI
about
here

The signs are stable across these alternative constructions. The expected-inflation component remains negatively related to yield spread changes, while the uncertainty component remains positively related to yield spread changes. The explanatory power is similar across measures. The baseline swap specification explains 27.0% of residual variation, compared with 25.7% using the 10-year swap rate, 28.8% using TIPS breakevens, and 19.9% using CPI-based proxies. The CPI specification is weaker, as expected, because realized inflation is more backward-looking, but it delivers the same qualitative conclusion.

I also control directly for lagged realized inflation. I do not include contemporaneous CPI inflation because it is released with a delay and is not in the investor information set when prices are measured. Lagged CPI inflation, however, captures recently observed inflation conditions that may affect spreads through backward-looking updating or through macroeconomic conditions correlated with inflation realizations. Including lagged CPI inflation leaves the main results essentially unchanged.

Cyclical proxy admits a second, quite different measurement. The baseline cyclical proxy is the rolling correlation between changes in expected inflation and revisions in 12-month-ahead expected real GDP growth from Consensus Economics. This measure is designed to capture

whether inflation news arrives together with stronger or weaker expected real activity. As a reduced-form alternative, I use the stock-bond return correlation, which is closely related to the cyclical nature of inflation risk and the hedge value of nominal bonds. I compute this measure as the 90-trading-day rolling correlation between the daily return on a synthetic 10-year zero-coupon Treasury and the CRSP value-weighted market return, and define $\Delta\rho_t^{SB}$ as its monthly change. A higher stock-bond correlation indicates that nominal bonds provide less hedge value in bad real states, consistent with a greater role for inflation shocks that are adverse for real activity. I therefore expect increases in this proxy to be associated with wider credit spreads. Column (6) of Table VI bears this out: $\Delta\rho_t^{SB}$ enters positively and significantly, and the expected-inflation and inflation-uncertainty coefficients are essentially unchanged.

Finally, I examine whether the baseline uncertainty proxy is related to disagreement among forecasters. The baseline measure, $\Delta\sigma_{i,t}^S$, is based on time-series variation in the inflation swap curve at the bond’s horizon. Forecast disagreement captures a different concept because it measures dispersion in beliefs across forecasters. To compare the two measures, I augment the baseline specification with changes in Consensus Economics forecast disagreement. Forecast disagreement enters with the expected positive sign and is statistically significant. More importantly, the inflation-swap volatility proxy remains positive after controlling for disagreement. The same conclusion holds when I replace swap volatility with the component orthogonal to forecast disagreement. The baseline uncertainty result therefore contains information beyond cross-sectional disagreement among forecasters.

Overall, the results show that the main findings are not tied to one inflation-risk measure and remain robust to controls for realized inflation and forecast disagreement.

B. Aggregate Conditions and Common Factors

The next set of tests asks how the inflation-risk measures relate to broader macroeconomic and monetary-policy conditions. This is especially relevant for the common-factor evidence,

since inflation risk moves together with real activity, monetary policy, and aggregate risk.

Table VII starts from the bond-level specification and adds variables that may account for related aggregate variation. Each column pair contrasts a specification with the additional controls alone against one that also includes the inflation-risk proxies. I first include the inflation volatility risk factor of Ceballos (2021); I then add real consumption growth, income growth, and changes in unemployment, followed by monetary-policy variables such as changes in the federal funds rate and the monetary-policy uncertainty measure of Husted, Rogers and Sun (2020). Expected inflation and inflation uncertainty keep their signs and significance across these specifications, though the expected-inflation coefficient attenuates once the real-activity controls are added, while inflation cyclicalities weaken to marginal significance once the monetary-policy variables and firm-level equity volatility enter. Their explanatory power remains close to the baseline.

Table
VII
about
here

The same table also adds firm-level equity volatility, following Campbell and Taksler (2003). Equity volatility enters with the expected positive sign, while the inflation-risk coefficients remain economically meaningful. The results are therefore not simply capturing the standard firm-level volatility channel in corporate credit.

Table VII also addresses two measurement concerns about the inflation-risk proxies. Because corporate yield spreads subtract a cash-flow-matched nominal Treasury yield, I first control directly for the corresponding duration-matched TIPS real-yield change, $\Delta \text{Real}_{i,t}$, and the change in its rolling 21-day volatility, $\Delta \sigma_{i,t}^{\text{Real}}$. Column (9) shows that expected inflation and inflation uncertainty remain significant and somewhat larger in magnitude, while the combined inflation and real-yield block explains 35.2% of residual variation, against 27.0% for the three inflation proxies alone. The real-yield change is itself significant, while real-yield volatility is not. Inflation cyclicalities, by contrast, are small and insignificant in this specification. Thus, movements in real yields do not absorb the inflation-risk results.

A second concern is that the baseline uncertainty measure, which is based on the 21-day rolling standard deviation of the level of the duration-matched swap rate, may partly capture

smooth movements in expected inflation over the month. I therefore reconstruct uncertainty from daily first differences of the swap rate, yielding $\Delta\sigma_{i,t}^{S,dd}$. Column (10) shows that its coefficient remains positive and highly significant, while the fraction of residual variation explained remains similar at 25.3%. The two coefficients are not directly comparable in magnitude, since one scales the standard deviation of daily changes and the other that of daily levels; what matters is that the drift-robust version delivers the same sign, significance, and explanatory power. The uncertainty result therefore does not appear to be driven by within-month drift in expected inflation.²⁷

C. Aggregate Conditions and the Residual Common Factor

The comparison so far has been at the bond level. Table VIII turns to the common component itself and compares the explanatory power of the inflation-risk proxies with that of a broader set of macroeconomic, monetary-policy, and growth-expectation variables.

The broader set of aggregate variables explains at least as much of the residual variation and its common component as the inflation-risk proxies. Using the CDGM controls, inflation risk accounts for 36.6% of residual variation and, in adjusted R^2 , for 21.2% of PC1, compared with 43.3% and 22.4% for the macroeconomic variables. With the full set of credit-risk and intermediation controls, the corresponding figures are 27.0% and 5.8% for inflation risk and 31.7% and 6.0% for the broader set of aggregate variables. Thus, inflation risk captures an important part of the aggregate variation in corporate yield spread changes, but the common factor is not solely an inflation phenomenon.

Table
VIII
about
here

To isolate the part of the inflation-risk proxies that is not shared with these aggregate conditions, I orthogonalize each proxy with respect to the macroeconomic, monetary-policy, and growth-expectation variables. The orthogonalized proxies continue to explain residual variation, accounting for 18.8% with the CDGM controls and 20.0% with the full controls.

²⁷The correlation between the level of duration-matched swap-rate volatility and the absolute monthly change in expected inflation is 0.53 across months; the correlation between the baseline proxy, which is the monthly change in that volatility, and the same absolute change is 0.49 in the bond-month panel.

They also explain 5.8% and 2.9%, respectively, of PC1 based on adjusted R^2 . This provides a more stringent measure of the variation associated specifically with inflation risk, net of its relation with broader aggregate conditions.

I then combine the orthogonalized inflation-risk proxies with the macroeconomic, monetary-policy, and growth-expectation variables in a single horse-race specification. The combined model explains 54.6% of residual variation with the [CDGM](#) controls and 44.9% with the full controls, with corresponding PC1 values of 28.7% and 9.1%. The increase relative to either set of variables alone shows that inflation risk does not exhaust the residual common component, but explains a separate part of it.

D. Alternative Residual Groupings and Extended Sample

Because the common factor is extracted from residuals averaged within maturity-leverage portfolios, I test whether the results are sensitive to the [CDGM](#) grouping choice.

Table [IX](#) repeats the exercise using alternative residual groupings. I keep maturity as one sorting dimension and pair it with leverage, rating, trading volume, market beta, and VIX beta. For each grouping, I compute cohort-month average residuals and extract principal components from the covariance matrix. Across these specifications, inflation risk explains between 23.8% and 29.1% of residual variation. This range is close to the 27.0% estimate from the baseline maturity-leverage grouping.

Table
[IX](#)
about
here

The effect on the residual common component is also stable. Without inflation risk, PC1 explains between 60.5% and 74.9% of residual variation, depending on the grouping. After adding the inflation-risk proxies, the PC1 share falls in every case, by between 4.9 and 5.6 percentage points. The common-factor evidence is therefore not driven by the specific way in which the residual portfolios are formed.

I next extend the sample through December 2024 using Enhanced TRACE. This extension adds the post-pandemic inflation surge and the Federal Reserve’s rapid tightening cycle, a period in which inflation risk became much more salient for firms and financial markets. The

post-2021 analysis is best viewed as a robustness check rather than a like-for-like extension of the baseline, because Enhanced TRACE lacks the dealer identifiers needed to construct the full set of OTC market proxies, so the extended specifications use a restricted intermediation-control set.²⁸

Table X reports the extended-sample results. The signs are unchanged and the explanatory power is close to the baseline. In the fully controlled extended sample, the inflation-risk block explains 23.1% of residual variation, compared with 27.0% in the baseline sample. The PC1 regressions also continue to show that inflation risk is related to the common residual component, with a PC1 Adjusted R^2 of 9.1%. Panel D reaches the same conclusion in fixed-effects regressions, where the signs of all three inflation-risk proxies are unchanged when the model is estimated on the extended panel. Inflation cyclicalities are also stronger and more statistically significant in the extended sample, which adds several years of the post-COVID inflation and monetary-tightening episode.

Table X
about
here

E. Time Variation in Inflation Sensitivity

The pooled estimates average over very different inflation regimes. This is particularly relevant for inflation cyclicalities, whose effect should depend on the macroeconomic state, but it may also matter for expected inflation and inflation uncertainty if these channels become more important when inflation risk is more salient.

Figure 2 plots 24-month rolling estimates of the coefficients on the three inflation-risk proxies. The expected-inflation coefficient is close to zero before the Global Financial Crisis, becomes sharply negative during the crisis, and remains mostly negative afterward, with its largest negative magnitudes occurring during the crisis and again around the post-COVID inflation surge. This pattern is consistent with the nominal-debt channel becoming more important when changes in the inflation outlook have larger effects on the real burden of

Figure 2
about
here

²⁸The Enhanced TRACE database does not include dealer identifiers. In the extended-sample tests, I use the subset of OTC market proxies that can be constructed without them: changes in dealer inventory, amount outstanding, block trades, the TED spread, and aggregate ratings. These variables represent the majority of the effect documented in [Friedwald and Nagler \(2019\)](#).

nominal liabilities.

Inflation uncertainty follows a less monotone path. Its coefficient is modestly positive before the crisis, turns negative during the crisis and again in 2017–2018, and rises sharply from 2019 onward to reach its largest values in the post-COVID period – precisely when the inflation outlook was hardest to forecast, and when uncertainty about nominal firm value should matter most for the default option.

Inflation cyclicalities behave differently. Its coefficient varies more over time, becoming positive during parts of the Global Financial Crisis and other episodes but remaining close to zero or changing sign elsewhere. This is consistent with a state-dependent channel whose effect depends on whether inflation news is associated with stronger or weaker expected real activity.

The shaded episodes in Figure 2 illustrate these differences. During the Global Financial Crisis, expected inflation becomes more negatively related to spreads and cyclicalities turn positive. The 2014–2015 oil-price disinflation episode is milder and only partly parallel: cyclicalities turn positive again, but the expected-inflation coefficient moves toward zero rather than becoming more negative. In the post-COVID period, expected inflation remains negatively related to spreads while the uncertainty coefficient becomes substantially larger.

Overall, the time-series evidence shows that sensitivity to expected inflation strengthens when inflation risk is more salient, while sensitivity to inflation uncertainty rises when the inflation outlook is harder to forecast. Cyclicalities are weaker on average and vary more across macroeconomic states, which helps explain their less stable pooled coefficient.²⁹

VI. Conclusion

This paper shows that inflation risk explains a substantial part of the variation in corporate yield spread changes left unexplained by standard credit-risk and intermediation variables.

²⁹Subperiod regressions in the Internet Appendix show similar patterns: the expected-inflation and inflation-uncertainty coefficients are insignificant over 2004–2007 and largest over 2020–2021, while inflation cyclicalities are significantly positive over 2008–2019 and turn negative over 2020–2021.

Expected inflation, inflation uncertainty, and inflation cyclicalities account for a sizeable share of this residual variation, including its common component across bonds. Higher expected inflation is associated with narrower spreads, while greater inflation uncertainty is associated with wider spreads.

The results also show that inflation-risk sensitivity varies systematically over time and across firms. Expected-inflation sensitivity becomes substantially stronger during periods of macroeconomic stress, while inflation-uncertainty sensitivity rises sharply after 2019. In the cross-section, sensitivities are stronger for firms with weaker fundamentals, lower credit quality, and higher leverage, and weaker when liabilities adjust more quickly through refinancing or floating-rate borrowing. Operating exposure provides an additional source of heterogeneity, with stronger expected-inflation sensitivity among low-markup firms and stronger sensitivity to both expected inflation and inflation uncertainty among firms with greater price pass-through. The variance decomposition shows that these differences in sensitivity do not map one-for-one into differences in explanatory power. Expected inflation makes the largest contribution across the cross-sectional sorts, while the operating characteristics generate more pronounced differences in sensitivity than in explained variation, and inflation cyclicalities play a smaller and less systematic role.

Taken together, the common residual component is systematically related to inflation risk, while the cross-sectional patterns are consistent with the channels through which inflation affects corporate credit risk. The evidence is consistent with structural models that combine nominal debt, slow leverage adjustment, and inflation dynamics. More broadly, the findings establish inflation risk as an important source of variation in corporate credit spreads, particularly when changes in inflation affect the real value and riskiness of promised payments.

Figure 1: Which Dimension of Inflation Risk Explains Residual Spread Variation

For each heterogeneity dimension, bonds are assigned to groups each month using the value of the relevant characteristic available strictly before the observation, following the group construction in Table V. I first estimate the control specification separately for each bond over its full sample, including the CDGM structural credit-risk variables and the intermediation controls from FN, HKS, and EHL. I then average the resulting residuals within each group and month and compute the fraction of variation explained (FVE) by adding the inflation-risk proxies ($\Delta E[\mu^S]_{i,t}$, $\Delta \sigma^S_{i,t}$, $\Delta \rho^S_t$) introduced in Section III. Each proxy is added separately, and its contribution is measured as the reduction in group-level residual variance relative to a common residual-variance denominator within each heterogeneity dimension. The stacked bars report the individual contributions of expected inflation, inflation uncertainty, and inflation cyclicalty, while the diamonds report the joint FVE from adding all three proxies. Because the proxies are correlated, the individual contributions need not sum to the joint FVE. Groups within each panel are ordered from low to high. The sample is based on U.S. industrial corporate bond transaction data from TRACE for the period 2004–2021.

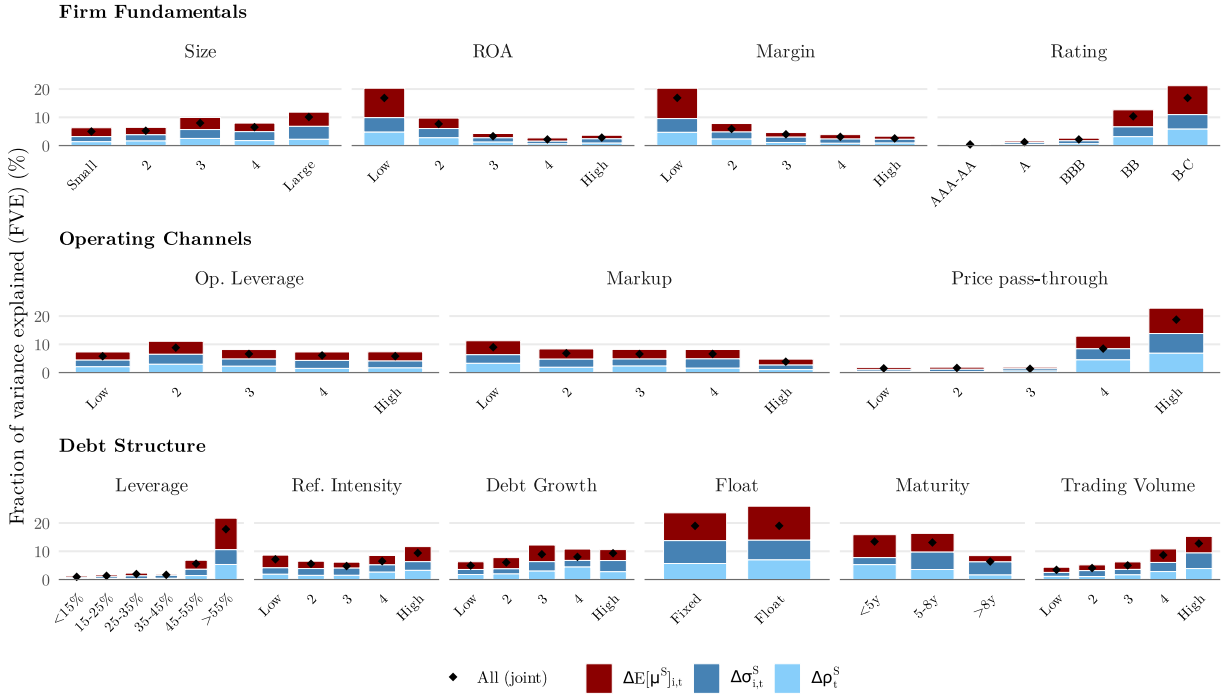


Figure 2: **Rolling Estimates of Inflation Risk Sensitivities**

This figure reports rolling estimates of the sensitivity of yield spread changes to the proxies of inflation risk. For each 24-month window, I estimate a pooled bond-fixed-effects regression, implemented by within-bond demeaning, on all bonds observed in that window. The model includes the [CDGM](#) structural credit-risk controls and the proxies of inflation risk ($\Delta E[\mu^S]_{i,t}$, $\Delta \sigma^S_{i,t}$, $\Delta \rho^S_t$) introduced in [Section III](#), together with the intermediation controls of [FN](#), [HKS](#), and [EHL](#). The left panel reports expected inflation, the middle panel reports inflation uncertainty, and the right panel reports inflation cyclical. Grey bands show 95% confidence intervals. Shaded regions denote the Global Financial Crisis, the oil-price disinflation episode, and the post-COVID inflation surge. The sample is based on U.S. industrial corporate bond transaction data from TRACE for the period 2004–2021.

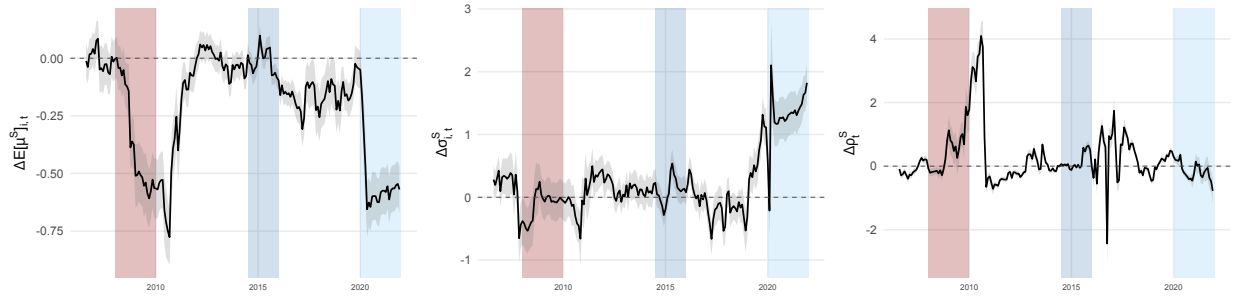


Table I: **Summary Statistics**

This table reports summary statistics of the data. Panel A reports the number of observations, mean, standard deviation, and 5%, 25%, 50%, 75% and 95% quantiles of bond characteristics, yield spreads ($YS_{i,t}$), and the proxies of inflation risk ($\Delta E[\mu^S]_{i,t}$, $\Delta \sigma_{i,t}^S$, $\Delta \rho_t^S$) introduced in Section III. The bond characteristics include the offering amount, the coupon rate, the bond age, the time to maturity, the duration, and the credit rating. The bond's rating is determined as the average of ratings provided by Standard & Poor's (S&P), Moody's, and Fitch when more than one are available or as the rating provided by one of the three rating agencies when only one rating is available. Panel B reports the standard deviation and the correlation matrix of changes in yield spreads and changes in the proxies of inflation risk. The sample is based on U.S. industrial corporate bond transaction data from TRACE for the period 2004–2021.

Panel A: Summary Statistics								
	Obs.	Mean	Std.	5%	25%	50%	75%	95%
Offering amount (mil.)	449788	741.84	658.34	200	350	500	1000	2000
Coupon (%)	449788	5.39	1.97	2.25	3.88	5.38	6.88	8.62
Age	449788	5.51	5.33	0.52	2.05	3.98	7.06	17.43
Time to Maturity	449788	8.99	8.08	0.96	3.38	6.15	10.00	27.27
Duration	449788	6.49	4.29	1.00	3.32	5.46	8.25	15.54
Rating	449788	8.75	3.47	4	6	9	10	15
Monthly Volume (mil.)	449788	53.75	100.88	0.69	7.06	24.20	62.65	198.27
Leverage	449788	0.31	0.21	0.07	0.16	0.26	0.42	0.75
Yield spread (%)	449788	2.32	3.05	0.32	0.81	1.44	2.71	6.57
$\Delta YS_{i,t}$	449788	0.74	82.76	-64.79	-14.89	-1.06	12.64	67.39
$E[\mu^S]_{i,t}$ (%)	449788	2.09	0.59	1.20	1.79	2.13	2.48	2.89
$\Delta E[\mu^S]_{i,t}$	449788	-0.18	25.12	-31.12	-9.69	0.55	11.67	30.73
$\Delta \sigma_{i,t}^S$	449788	0.07	7.27	-6.83	-2.03	-0.07	1.90	7.70
$\Delta \rho_t^S$	449788	-0.05	10.11	-14.05	-2.39	0.27	3.51	11.71
Panel B: Correlation Matrix								
	Std.	$\Delta YS_{i,t}$	$\Delta E[\mu^S]_{i,t}$	$\Delta \sigma_{i,t}^S$	$\Delta \rho_t^S$			
$\Delta YS_{i,t}$	0.828	1	-0.299	0.236	0.166			
$\Delta E[\mu^S]_{i,t}$	0.251		1	-0.346	-0.153			
$\Delta \sigma_{i,t}^S$	0.073			1	0.354			
$\Delta \rho_t^S$	0.101				1			

Table II: Inflation Risk and Yield Spread Changes

For each industrial bond i with at least 25 monthly observations of yield spread changes $\Delta YS_{i,t}$, I estimate the model

$$\Delta YS_{i,t} = \alpha_i + \beta_i^T \Delta S_{i,t} + \theta_i^T \Delta I_{i,t} + \Gamma_i^T \Delta C_{i,t} + v_{i,t},$$

where $\Delta S_{i,t}$ represents structural model variables, $\Delta I_{i,t}$ denotes the proxies of inflation risk ($\Delta E[\mu^S]_{i,t}$, $\Delta \sigma_{i,t}^S$, $\Delta \rho_i^S$) introduced in Section III, and $\Delta C_{i,t}$ refers to control proxies from [Friedwald and Nagler \(2019\)](#), [He, Khorrami and Song \(2022\)](#), and [Eisfeldt, Herskovic and Liu \(2024\)](#). Column (1) reports the CDGM regression; Columns (2)–(4) include each proxy individually; Column (5) includes all three jointly; Columns (6)–(8) progressively add controls. Panel A presents average coefficients, t-statistics, mean and median adjusted R^2 values, and sample sizes. Panel B details a principal component analysis on the residuals averaged within 18 cohorts (3 maturity $\times$ 6 leverage bins), reporting the variance explained by the first two principal components (PC1, PC2), the total unexplained variance (UV), and the fraction of variation explained (FVE) by adding the variables that enter each column. Panel C includes statistics from time-series regressions of the PC1 on the variables added in each column. The sample is based on U.S. industrial corporate bond transaction data from TRACE for the period 2004–2021.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Individual Bond Regressions								
$\Delta E[\mu^S]_{i,t}$		-0.535 (-37.582)			-0.478 (-35.756)	-0.380 (-25.155)	-0.393 (-24.900)	-0.326 (-20.845)
$\Delta \sigma_{i,t}^S$			1.600 (36.329)		1.278 (30.425)	1.123 (24.238)	1.165 (21.634)	0.714 (12.776)
$\Delta \rho_i^S$				0.604 (31.405)	0.252 (15.033)	0.078 (3.037)	0.064 (2.325)	0.058 (1.995)
CDGM	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
FN	No	No	No	No	No	Yes	Yes	Yes
HKS	No	No	No	No	No	No	Yes	Yes
EHL	No	No	No	No	No	No	No	Yes
Mean Adj. R^2	0.355	0.386	0.418	0.390	0.447	0.507	0.516	0.530
Median Adj. R^2	0.378	0.409	0.431	0.410	0.463	0.540	0.555	0.573
Obs.	449788	449788	449788	449788	449788	449788	449788	449788
Bonds	6826	6826	6826	6826	6826	6826	6826	6826
Panel B: Principal Component Analysis								
FVE		0.200	0.179	0.088	0.366	0.288	0.287	0.270
PC1	0.797	0.743	0.759	0.786	0.711	0.691	0.686	0.652
PC2	0.054	0.078	0.064	0.059	0.084	0.084	0.092	0.097
UV	1.124	0.900	0.923	1.025	0.713	0.441	0.392	0.319
Panel C: Time-Series Regression of PC1 on Added Variables								
Adj. R^2		0.129	0.152	0.063	0.212	0.113	0.098	0.058
R^2		0.133	0.156	0.067	0.223	0.126	0.111	0.072
F-stat		31.640	38.125	14.827	19.544	9.833	8.528	5.283
p -value		0.000	0.000	0.000	0.000	0.000	0.000	0.002
Obs.		208	208	208	208	208	208	208

Table III: Inflation Expectations and Yield Spread Changes

For each industrial bond i with at least 25 monthly observations of yield spread changes $\Delta YS_{i,t}$, I estimate

$$\Delta YS_{i,t} = \alpha_i + \beta_i^T \Delta S_{i,t} + \theta_i^T \Delta I_{i,t} + \Gamma_i^T \Delta C_{i,t} + v_{i,t},$$

where $\Delta S_{i,t}$ is the vector of CDGM structural credit-risk controls, $\Delta I_{i,t}$ contains survey-anchored proxies of inflation risk ($\Delta E[\mu^E]_{i,t}$, $\Delta \sigma_{i,t}^E$, $\Delta \rho_t^E$), and $\Delta C_{i,t}$ contains the intermediation controls from [Friewald and Nagler \(2019\)](#), [He, Khorrami and Song \(2022\)](#), and [Eisfeldt, Herskovic and Liu \(2024\)](#). The survey-anchored proxies are obtained from a Nelson-Siegel-Svensson curve fitted daily to inflation swaps, TIPS breakevens, and survey expectations, and evaluated at each bond's duration. Column (1) excludes the inflation-risk proxies. Columns (2), (3), and (5) add expected inflation, inflation uncertainty, and inflation cyclicalilty individually, column (4) adds expected inflation and inflation uncertainty together, and column (6) includes all three. All columns include the structural credit-risk and intermediation controls. Panel A presents average coefficients, t-statistics, mean and median adjusted R^2 values, and sample sizes. Panel B details a principal component analysis on the residuals averaged within 18 cohorts (3 maturity $\times$ 6 leverage bins), reporting the variance explained by the first two principal components (PC1, PC2), the total unexplained variance (UV), and the fraction of variation explained (FVE) by adding the variables that enter each column. Panel C includes statistics from time-series regressions of the PC1 on the variables added in each column. The sample is based on U.S. industrial corporate bond transaction data from TRACE for the period 2004–2021.

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Individual Bond Regressions						
$\Delta E[\mu^E]_{i,t}$		-0.255 (-18.755)		-0.292 (-20.078)		-0.274 (-18.472)
$\Delta \sigma_{i,t}^E$			0.522 (9.999)	0.422 (6.576)		0.364 (4.918)
$\Delta \rho_t^E$					0.168 (3.834)	0.180 (3.535)
CDGM	Yes	Yes	Yes	Yes	Yes	Yes
FN	Yes	Yes	Yes	Yes	Yes	Yes
HKS	Yes	Yes	Yes	Yes	Yes	Yes
EHL	Yes	Yes	Yes	Yes	Yes	Yes
Mean Adj. R^2	0.505	0.515	0.519	0.528	0.511	0.532
Median Adj. R^2	0.534	0.545	0.547	0.560	0.545	0.569
Obs.	449788	449788	449788	449788	449788	449788
Bonds	6826	6826	6826	6826	6826	6826
Panel B: Principal Component Analysis						
FVE		0.098	0.084	0.169	0.120	0.243
PC1	0.708	0.677	0.696	0.668	0.683	0.653
PC2	0.085	0.097	0.085	0.100	0.091	0.104
UV	0.437	0.394	0.400	0.363	0.384	0.330
Panel C: Time-Series Regression of PC1 on Added Variables						
Adj. R^2		0.039	0.033	0.047	0.010	0.043
R^2		0.043	0.038	0.056	0.015	0.057
F-stat		9.310	8.140	6.064	3.110	4.107
p-value		0.003	0.005	0.003	0.079	0.007
Obs.		208	208	208	208	208

Table IV: **Inflation Risk and Yield Spread Changes: Estimation Approach and Inference**

For each industrial bond i with at least 25 monthly observations of yield spread changes $\Delta YS_{i,t}$, I estimate the fully controlled specification including the CDGM structural credit-risk controls, the intermediation controls from [Friewald and Nagler \(2019\)](#), [He, Khorrami and Song \(2022\)](#), and [Eisfeldt, Herskovic and Liu \(2024\)](#), and the proxies of inflation risk ($\Delta E[\mu^S]_{i,t}$, $\Delta \sigma_{i,t}^S$, $\Delta \rho_t^S$) introduced in Section III. Column (1) reports mean-group estimates, obtained by estimating the model separately for each bond and averaging coefficients across bonds. The t-statistics in column (1) are computed from the cross-sectional dispersion of bond-level coefficient estimates, following CDGM. Columns (2)–(4) report panel fixed-effects estimates with bond fixed effects and homogeneous slopes. The columns differ only in the clustering of standard errors: column (2) double-clusters by bond and month, column (3) clusters by month, and column (4) clusters by bond. The sample is based on U.S. industrial corporate bond transaction data from TRACE for the period 2004–2021.

	Individual Bond Regressions	Panel FE		
	(CDGM)	Bond×Time	Time only	Bond only
	(1)	(2)	(3)	(4)
$\Delta E[\mu^S]_{i,t}$	-0.326*** (-20.84)	-0.339*** (-4.53)	-0.339*** (-4.53)	-0.339*** (-19.05)
$\Delta \sigma_{i,t}^S$	0.714*** (12.78)	0.587*** (2.86)	0.587*** (2.89)	0.587*** (8.42)
$\Delta \rho_t^S$	0.058** (2.00)	0.167 (1.40)	0.167 (1.40)	0.167*** (10.02)
CDGM	Yes	Yes	Yes	Yes
FN	Yes	Yes	Yes	Yes
HKS	Yes	Yes	Yes	Yes
EHL	Yes	Yes	Yes	Yes
Bond FE	Bond-level TS	Yes	Yes	Yes
Estimator	Mean Group	Panel FE	Panel FE	Panel FE
Clustering	Cross-bond	Bond×Time	Time only	Bond only
Observations	449788	449788	449788	449788

Table V: **Heterogeneity in Inflation-Risk Sensitivity**

For each heterogeneity dimension, I estimate a bond-fixed-effects panel regression including the [CDGM](#) structural credit-risk controls, the intermediation controls from [FN](#), [HKS](#), and [EHL](#), and the three inflation-risk proxies ($\Delta E[\mu^S]_{i,t}$, $\Delta \sigma_{i,t}^S$, $\Delta \rho_t^S$) introduced in Section III. The inflation-risk proxies are interacted with the full set of group indicators for each dimension, while the coefficients on the credit-risk and intermediation controls are common across groups. The table reports the High-Minus-Low difference in each inflation-risk coefficient, defined as the coefficient for the highest group minus that for the lowest group, together with two-way clustered t-statistics. Groups are formed using information available strictly before the yield spread observation. I use a one-month lag for the time-varying characteristics, except for size, which is fixed at issuance, and price pass-through and debt growth, which use expanding means of the underlying monthly series through twelve months before the observation.

Dimension	$\Delta E[\mu^S]_{i,t}$	$\Delta \sigma_{i,t}^S$	$\Delta \rho_t^S$	Obs.	Bonds
<i>Panel A: Firm Fundamentals</i>					
Size	0.242*** (3.24)	0.024 (0.08)	-0.370 (-1.50)	449788	6826
ROA	1.023*** (8.69)	-1.098** (-2.27)	-0.682*** (-2.71)	424180	6826
Margin	0.887*** (8.41)	-0.746* (-1.66)	-0.729*** (-3.18)	423975	6,823
Rating	-1.761*** (-9.37)	1.696* (1.82)	1.081* (1.88)	424180	6826
<i>Panel B: Operating Channels</i>					
Op. Leverage	0.052 (0.55)	0.255 (0.95)	0.031 (0.29)	424180	6826
Markup	0.296*** (3.99)	0.003 (0.01)	-0.163 (-1.20)	423975	6823
Price pass-through	-0.440*** (-3.47)	0.925* (1.79)	0.245 (0.79)	108686	2749
<i>Panel C: Debt Structure</i>					
Leverage	-1.394*** (-8.31)	1.492* (1.94)	1.136** (2.07)	424180	6826
Ref. Intensity	0.245** (2.49)	-0.694* (-1.66)	-0.227 (-1.54)	420299	6788
Debt Growth	-0.251** (-2.35)	0.169 (0.49)	-0.260 (-1.62)	293639	6824
Float	0.203*** (4.00)	-0.280* (-1.76)	0.082 (0.90)	424180	6826
Maturity	0.027 (0.28)	0.275 (0.66)	-0.736*** (-2.81)	424180	6826
Trading Volume	-0.165** (-2.47)	0.323 (1.37)	-0.035 (-0.22)	424180	6826

Table VI: **Alternative Inflation-Risk Proxies**

For each industrial bond i with at least 25 monthly observations of yield spread changes $\Delta YS_{i,t}$, I estimate

$$\Delta YS_{i,t} = \alpha_i + \beta_i^T \Delta S_{i,t} + \theta_i^T \Delta I_{i,t} + \Gamma_i^T \Delta C_{i,t} + v_{i,t},$$

where $\Delta S_{i,t}$ represents structural model variables, $\Delta I_{i,t}$ contains the proxies of inflation risk ($\Delta E[\mu^S]_{i,t}$, $\Delta \sigma_{i,t}^S$, $\Delta \rho_i^S$) introduced in Section III, and $\Delta C_{i,t}$ refers to control proxies from Friewald and Nagler (2019), He, Khorrami and Song (2022), and Eisfeldt, Herskovic and Liu (2024). All columns include $\Delta S_{i,t}$ and $\Delta C_{i,t}$. Column (1) excludes inflation-risk proxies. Columns (2)–(6) use alternative proxy sets: baseline duration-matched swaps, 10-year swaps, duration-matched TIPS breakevens, CPI-based proxies, and the stock–bond cyclicity proxy. Columns (7)–(10) add, respectively, lagged CPI inflation, lagged CPI inflation with the baseline swap proxies, forecast disagreement, and inflation uncertainty orthogonalized to forecast disagreement. Panel A presents average coefficients, t-statistics, mean and median adjusted R^2 values, and sample sizes. Panel B details a principal component analysis on the residuals averaged within 18 cohorts (3 maturity $\times$ 6 leverage bins), reporting the variance explained by the first two principal components (PC1, PC2), the total unexplained variance (UV), and the fraction of variation explained (FVE) by adding the variables that enter each column. Panel C includes statistics from time-series regressions of the PC1 on the variables added in each column. The sample is based on U.S. industrial corporate bond transaction data from TRACE for the period 2004–2021.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Panel A: Individual Bond Regressions										
Proxies		Swaps	Swaps 10Y	TIPS	CPI	Swaps	CPI lag	Swaps	Swaps orth.	
$\Delta E[\mu^S]_{i,t}$		-0.326 (-20.845)	-0.316 (-13.014)	-0.314 (-21.777)	-0.120 (-11.410)	-0.350 (-22.199)		-0.325 (-12.004)	-0.310 (-17.504)	-0.307 (-19.867)
$\Delta \sigma_{i,t}^S$		0.714 (12.776)	0.861 (9.265)	0.416 (7.573)	0.224 (6.010)	0.693 (10.782)		0.760 (12.368)	0.796 (10.696)	0.708 (12.449)
$\Delta \rho_i^S$		0.058 (1.995)	0.024 (0.876)	0.204 (5.788)	0.175 (3.385)			0.097 (3.229)	0.095 (2.126)	0.091 (3.084)
$\Delta \rho_i^{SB}$						0.107 (5.414)				
CPI lag							6.720 (9.694)	6.562 (7.468)		
CE disag.									0.173 (2.730)	
CDGM	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
FN	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
HKS	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
EHL	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean Adj. R^2	0.505	0.530	0.525	0.529	0.511	0.527	0.509	0.533	0.534	0.530
Median Adj. R^2	0.534	0.573	0.569	0.569	0.552	0.564	0.542	0.582	0.577	0.571
Obs.	449788	449788	449788	449788	449788	449788	449788	449788	449788	449788
Bonds	6826	6826	6826	6826	6826	6826	6826	6826	6826	6826
Panel B: Principal Component Analysis										
FVE		0.270	0.257	0.288	0.199	0.262	0.075	0.298	0.302	0.273
PC1	0.708	0.652	0.655	0.644	0.670	0.655	0.695	0.650	0.651	0.652
PC2	0.085	0.097	0.095	0.100	0.096	0.098	0.087	0.097	0.096	0.095
UV	0.437	0.319	0.324	0.311	0.350	0.322	0.404	0.306	0.305	0.317
Panel C: Time-Series Regression of PC1 on Added Variables										
Adj. R^2		0.058	0.095	0.107	0.058	0.068	0.027	0.084	0.066	0.062
R^2		0.072	0.108	0.120	0.072	0.081	0.031	0.101	0.084	0.075
F-stat		5.283	8.254	9.295	5.240	6.017	6.650	5.715	4.668	5.541
p -value		0.002	0.000	0.000	0.002	0.001	0.011	0.000	0.001	0.001
Obs.		208	208	208	208	208	208	208	208	208

Table VII: **Additional Variables**

For each industrial bond i with at least 25 monthly observations of yield spread changes $\Delta YS_{i,t}$, I estimate

$$\Delta YS_{i,t} = \alpha_i + \beta_i^T \Delta S_{i,t} + \theta_i^T \Delta I_{i,t} + \Gamma_i^T \Delta C_{i,t} + v_{i,t},$$

where $\Delta S_{i,t}$ represents structural model variables, $\Delta I_{i,t}$ contains the proxies of inflation risk ($\Delta E[\mu^S]_{i,t}$, $\Delta \sigma_{i,t}^S$, $\Delta \rho_i^S$) introduced in Section III, and $\Delta C_{i,t}$ refers to control proxies from [Friewald and Nagler \(2019\)](#), [He, Khorrami and Song \(2022\)](#), and [Eisfeldt, Herskovic and Liu \(2024\)](#). Columns (1)–(8) add auxiliary controls in pairs, with the inflation-risk proxies included in the even-numbered columns: inflation volatility risk (1)–(2), consumption growth, income growth, and unemployment (3)–(4), monetary-policy uncertainty and the federal funds rate (5)–(6), and firm-level equity volatility (7)–(8). Columns (9)–(10) start from the fully controlled baseline; column (9) adds the duration-matched real-yield change and its rolling 21-day volatility, while column (10) uses the drift-robust uncertainty proxy. Panel A presents average coefficients, t-statistics, mean and median adjusted R^2 values, and sample sizes. Panel B details a principal component analysis on the residuals averaged within 18 cohorts (3 maturity $\times$ 6 leverage bins), reporting the variance explained by the first two principal components (PC1, PC2), the total unexplained variance (UV), and the fraction of variation explained (FVE) by adding the variables that enter each column. Panel C includes statistics from time-series regressions of the PC1 on the variables added in each column. The sample is based on U.S. industrial corporate bond transaction data from TRACE for the period 2004–2021.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Panel A: Individual Bond Regressions										
ΔIVR_t	0.261 (8.01)	0.241 (3.66)								
$\Delta Consumption_t$			3.214 (2.96)	2.972 (2.79)						
$\Delta Income_t$			3.077 (5.47)	2.876 (3.98)						
$\Delta Unemployment_t$			0.088 (4.44)	0.059 (3.19)						
ΔMPU_t					-0.004 (-0.75)	0.000 (0.01)				
$\Delta FED Fund_t$					0.003 (0.09)	0.105 (2.65)				
$\Delta \sigma_{i,t}^{Equity}$							0.601 (6.01)	0.274 (2.25)		
$\Delta Real_{i,t}$									-0.135 (-4.55)	
$\Delta \sigma_{i,t}^{Real}$									-0.010 (-0.17)	
$\Delta \sigma_{i,t}^{S,dd}$										1.793 (10.88)
$\Delta E[\mu^S]_{i,t}$		-0.319 (-15.14)		-0.217 (-12.23)		-0.319 (-18.90)		-0.311 (-15.96)	-0.448 (-13.43)	-0.277 (-16.93)
$\Delta \sigma_{i,t}^S$		0.633 (9.37)		0.785 (9.84)		0.799 (11.06)		0.740 (10.68)	0.838 (11.74)	
$\Delta \rho_i^S$		0.101 (2.68)		0.080 (2.63)		0.057 (1.95)		0.057 (1.67)	-0.025 (-0.71)	0.116 (4.04)
Mean Adj. R^2	0.505	0.530	0.524	0.542	0.511	0.536	0.506	0.529	0.548	0.533
Median Adj. R^2	0.536	0.577	0.559	0.585	0.541	0.577	0.536	0.574	0.590	0.569
Obs.	449788	449788	449788	449788	449788	449788	449788	449788	449788	449788
Bonds	6826	6826	6826	6826	6826	6826	6826	6826	6826	6826
Panel B: Principal Component Analysis										
FVE		0.270		0.255		0.280	0.059	0.305	0.352	0.253
PC1	0.705	0.650	0.687	0.630	0.706	0.645	0.707	0.654	0.633	0.652
PC2	0.087	0.099	0.081	0.097	0.083	0.097	0.083	0.096	0.109	0.099
UV	0.415	0.303	0.351	0.261	0.407	0.293	0.411	0.304	0.283	0.326
Panel C: Time-Series Regression of PC1 on Added Variables										
Adj. R^2		0.060		0.055		0.057	0.007	0.055	0.110	0.055
R^2		0.074		0.068		0.071	0.011	0.073	0.131	0.069
F -stat		5.436		4.988		5.177	2.385	3.987	6.114	5.028
p -value		0.001		0.002		0.002	0.124	0.004	0.000	0.002
Obs.		208		208		208	208	208	208	208

Table VIII: Macro Conditions and the Residual Common Component

This table compares the explanatory power of inflation risk with macroeconomic, monetary-policy, and growth-expectation variables for the residual common component in corporate yield spread changes. The residuals are formed from bond-level spread-change regressions and averaged within maturity–leverage cohorts. The table reports results under two benchmark specifications: the CDGM structural credit-risk controls and the fully controlled specification that also includes the intermediation controls from [Friewald and Nagler \(2019\)](#), [He, Khorrami and Song \(2022\)](#), and [Eisfeldt, Herskovic and Liu \(2024\)](#). Columns compare four regressor blocks: the baseline inflation-risk proxies, macroeconomic, monetary-policy, and growth-expectation variables, inflation-risk proxies orthogonalized with respect to aggregate conditions, and a horse-race specification combining the orthogonalized inflation-risk proxies with the macroeconomic, monetary-policy, and growth-expectation variables. Panel A reports principal component statistics for the cohort residuals, including PC1, PC2, total unexplained variance (UV), and the fraction of variation explained (FVE) by adding the relevant block. Panel B reports statistics from time-series regressions of the residual PC1 on the corresponding block variables. The sample is based on U.S. industrial corporate bond transaction data from TRACE for the period 2004–2021.

	Inflation (1)	Macro+MP+Growth (2)	Infl. [⊥] (3)	Horse race (4)
CDGM				
Panel A: Principal Component Analysis				
FVE	0.366	0.433	0.188	0.546
PC1	0.711	0.718	0.771	0.683
PC2	0.084	0.073	0.058	0.080
UV	0.713	0.638	0.913	0.510
Panel B: Time-Series Regression of PC1 on Inflation-Risk or Macro Proxies				
Adj. R^2	0.212	0.224	0.058	0.287
R^2	0.223	0.246	0.071	0.318
F -stat	19.544	10.944	5.219	10.236
p -value	0.000	0.000	0.002	0.000
Obs.	208	208	208	208
CDGM + FN + HKS + EHL				
Panel A: Principal Component Analysis				
FVE	0.270	0.317	0.200	0.449
PC1	0.652	0.672	0.679	0.629
PC2	0.097	0.086	0.090	0.096
UV	0.319	0.298	0.350	0.241
Panel B: Time-Series Regression of PC1 on Inflation-Risk or Macro Proxies				
Adj. R^2	0.058	0.060	0.029	0.091
R^2	0.072	0.087	0.044	0.131
F -stat	5.283	3.206	3.097	3.314
p -value	0.002	0.005	0.028	0.001
Obs.	208	208	208	208

Table IX: **Different Residual Groups**

For each industrial bond i with at least 25 monthly observations of yield spread changes $\Delta YS_{i,t}$, I estimate the model:

$$\Delta YS_{i,t} = \alpha_i + \beta_i^T \Delta S_{i,t} + \theta_i^T \Delta I_{i,t} + \Gamma_i^T \Delta C_{i,t} + v_{i,t},$$

where $\Delta S_{i,t}$ represents structural model variables, $\Delta I_{i,t}$ contains the proxies of inflation risk ($\Delta E[\mu^S]_{i,t}$, $\Delta \sigma_{i,t}^S$, $\Delta \rho_i^S$) introduced in Section III, and $\Delta C_{i,t}$ refers to control proxies from Friewald and Nagler (2019), He, Khorrami and Song (2022), and Eisfeldt, Herskovic and Liu (2024). I then assign monthly residuals to cohorts using maturity as the first sorting variable and, alternatively, leverage, credit rating, trading volume, market beta, or VIX beta as the second sorting variable. For each grouping, I compute average cohort-month residuals and extract principal components from their covariance matrix. The table reports the share of residual variation explained by the first and second principal components (PC1 and PC2), separately for specifications without and with the inflation-risk proxies, and the fraction of variation explained (FVE) by adding the inflation-risk proxies. The sample is based on U.S. industrial corporate bond transaction data from TRACE for the period 2004–2021.

Group 1	Group 2	Inflation Risk	PC1	PC2	FVE
Time to Maturity	Leverage	No	0.708	0.085	0.270
		Yes	0.652	0.097	
Time to Maturity	Rating	No	0.659	0.113	0.268
		Yes	0.605	0.126	
Time to Maturity	Volume	No	0.749	0.075	0.291
		Yes	0.697	0.089	
Time to Maturity	Market Beta	No	0.640	0.090	0.251
		Yes	0.591	0.099	
Time to Maturity	VIX Beta	No	0.605	0.105	0.238
		Yes	0.551	0.116	

Table X: **Extended Sample**

For each industrial bond i with at least 25 monthly observations of yield spread changes $\Delta YS_{i,t}$, I estimate

$$\Delta YS_{i,t} = \alpha_i + \beta_i^T \Delta S_{i,t} + \theta_i^T \Delta I_{i,t} + \Gamma_i^T \Delta C_{i,t} + v_{i,t},$$

where $\Delta S_{i,t}$ is the vector of CDGM structural credit-risk controls, $\Delta I_{i,t}$ contains the proxies of inflation risk ($\Delta E[\mu^S]_{i,t}$, $\Delta \sigma_{i,t}^S$, $\Delta \rho_t^S$) introduced in Section III, and $\Delta C_{i,t}$ contains the available intermediation controls. Column (1) excludes the inflation-risk proxies. Columns (2)–(4) include each proxy individually. Column (5) includes all three jointly. Columns (6)–(8) progressively add the available controls. Panel A presents average coefficients, t-statistics, mean and median adjusted R^2 values, and sample sizes. Panel B details a principal component analysis on the residuals averaged within 18 cohorts (3 maturity $\times$ 6 leverage bins), reporting the variance explained by the first two principal components (PC1, PC2), the total unexplained variance (UV), and the fraction of variation explained (FVE) by adding the variables that enter each column. Panel C includes statistics from time-series regressions of the PC1 on the variables added in each column. Panel D reports panel fixed-effects estimates with standard errors double-clustered by bond and month. The sample is based on U.S. industrial corporate bond transaction data from Enhanced TRACE for the period 2004–2024.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Individual Bond Regressions								
$\Delta E[\mu^S]_{i,t}$		-0.325 (-32.812)			-0.296 (-31.381)	-0.278 (-29.054)	-0.262 (-27.065)	-0.228 (-23.833)
$\Delta \sigma_{i,t}^S$			1.338 (39.930)		1.108 (34.736)	0.988 (31.642)	0.886 (28.256)	0.524 (16.510)
$\Delta \rho_t^S$				0.451 (29.292)	0.245 (17.724)	0.221 (14.163)	0.222 (14.024)	0.137 (8.608)
CDGM	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
FN	No	No	No	No	No	Yes	Yes	Yes
HKS	No	No	No	No	No	No	Yes	Yes
EHL	No	No	No	No	No	No	No	Yes
Mean Adj. R^2	0.325	0.344	0.374	0.347	0.394	0.437	0.448	0.464
Median Adj. R^2	0.339	0.357	0.386	0.361	0.404	0.454	0.467	0.481
Obs.	580208	580208	580208	580208	580208	580208	580208	580208
Bonds	8327	8327	8327	8327	8327	8327	8327	8327
Panel B: Principal Component Analysis								
FVE		0.154	0.168	0.073	0.309	0.245	0.233	0.231
PC1	0.793	0.753	0.742	0.781	0.703	0.701	0.691	0.670
PC2	0.067	0.073	0.080	0.073	0.090	0.091	0.093	0.099
UV	0.745	0.631	0.620	0.691	0.515	0.427	0.394	0.338
Panel C: Time-Series Regression of PC1 on Added Variables								
Adj. R^2		0.107	0.170	0.048	0.213	0.143	0.123	0.091
R^2		0.111	0.174	0.052	0.223	0.153	0.134	0.103
F-stat		30.146	50.824	13.358	22.898	14.499	12.337	9.137
p -value		0.000	0.000	0.000	0.000	0.000	0.000	0.000
Obs		244	244	244	244	244	244	244
Panel D: Fixed Effects Regressions								
$\Delta E[\mu^S]_{i,t}$		-0.421*** (-3.432)			-0.345*** (-4.047)	-0.335*** (-3.810)	-0.303*** (-3.767)	-0.289*** (-3.890)
$\Delta \sigma_{i,t}^S$			1.488*** (3.522)		1.129*** (3.870)	1.001*** (3.913)	0.928*** (3.611)	0.720*** (3.235)
$\Delta \rho_t^S$				0.594** (2.431)	0.390** (2.565)	0.308** (2.534)	0.314*** (2.780)	0.204** (1.998)
CDGM	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
FN	No	No	No	No	No	Yes	Yes	Yes
HKS	No	No	No	No	No	No	Yes	Yes
EHL	No	No	No	No	No	No	No	Yes
Adj. R^2	0.147	0.162	0.164	0.153	0.176	0.180	0.185	0.191
R^2	0.159	0.174	0.176	0.165	0.188	0.192	0.197	0.203

REFERENCES

- Andreasen, M.M., Christensen, J.H.E., Riddell, S., 2021. The TIPS liquidity premium. *Review of Finance* 25, 1639–1675. URL: <https://academic.oup.com/rof/article/25/6/1639/6410663>, doi:<https://doi.org/10.1093/rof/rfab018>.
- Bahaj, S., Czech, R., Ding, S., Reis, R., 2023. The market for inflation risk. Staff Working Paper, 1028, Bank of England URL: <https://www.bankofengland.co.uk/working-paper/2023/the-market-for-inflation-risk>.
- Baldwin, C.Y., Ruback, R.S., 1986. Inflation, uncertainty, and investment. *The Journal of Finance* 41, 657–668. URL: <https://onlinelibrary.wiley.com/doi/10.1111/j.1540-6261.1986.tb04528.x>, doi:<https://doi.org/10.2307/2328496>.
- Bessembinder, H., Kahle, K.M., Maxwell, W.F., Xu, D., 2008. Measuring abnormal bond performance. *Review of Financial Studies* 22, 4219–4258. URL: <http://dx.doi.org/10.1093/rfs/hhn105>, doi:[10.1093/rfs/hhn105](https://doi.org/10.1093/rfs/hhn105).
- Bhamra, H.S., Dorion, C., Jeanneret, A., Weber, M., 2022. High inflation: Low default risk and low equity valuations. *The Review of Financial Studies* 36, 1192–1252. URL: <http://dx.doi.org/10.1093/rfs/hhac021>, doi:[10.1093/rfs/hhac021](https://doi.org/10.1093/rfs/hhac021).
- Bhamra, H.S., Fisher, A.J., Kuehn, L.A., 2011. Monetary policy and corporate default. *Journal of Monetary Economics* 58, 480–494. URL: <http://dx.doi.org/10.1016/j.jmoneco.2011.05.010>, doi:[10.1016/j.jmoneco.2011.05.010](https://doi.org/10.1016/j.jmoneco.2011.05.010).
- Bonelli, D., Palazzo, B., Yamarchy, R., 2025. Good inflation, bad inflation, and the dynamics of credit risk. Working Paper URL: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4798269.
- Boons, M., Duarte, F., de Roon, F., Szymanowska, M., 2020. Time-varying inflation risk and stock returns. *Journal of Financial Economics* 136, 444–470. URL: <https://www>.

[sciencedirect.com/science/article/abs/pii/S0304405X19302429](https://www.sciencedirect.com/science/article/abs/pii/S0304405X19302429), doi:10.1016/j.jfineco.2019.10.004.

Campbell, J.Y., Pflueger, C., Viceira, L.M., 2020. Macroeconomic drivers of bond and equity risks. *Journal of Political Economy* 128, 3148–3185. URL: <http://dx.doi.org/10.1086/707766>, doi:10.1086/707766.

Campbell, J.Y., Shiller, R.J., Viceira, L.M., 2012. Understanding inflation-indexed bond markets , 29–70 URL: <http://dx.doi.org/10.7551/mitpress/9037.003.0005>, doi:10.7551/mitpress/9037.003.0005.

Campbell, J.Y., Taksler, G.B., 2003. Equity volatility and corporate bond yields. *Journal of Finance* 58, 2321–2349. doi:10.1046/j.1540-6261.2003.00607.x.

Ceballos, L., 2021. Inflation volatility risk and the cross-section of corporate bond returns. Working Paper URL: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3883556.

Chen, H., 2010. Macroeconomic conditions and the puzzles of credit spreads and capital structure. *The Journal of Finance* 65, 2171–2212. URL: <http://dx.doi.org/10.1111/j.1540-6261.2010.01613.x>, doi:10.1111/j.1540-6261.2010.01613.x.

Chen, N.F., Roll, R., Ross, S.A., 1986. Economic forces and the stock market. *The Journal of Business* 59, 383. URL: <http://dx.doi.org/10.1086/296344>, doi:10.1086/296344.

Christensen, J.H.E., Lopez, J.A., Rudebusch, G.D., 2015. Pricing deflation risk with US treasury yields. *Review of Finance* 20, 1107–1152. URL: <http://dx.doi.org/10.1093/rof/rfv029>, doi:10.1093/rof/rfv029.

Collin-Dufresne, P., Goldstein, R.S., Martin, J.S., 2001. The determinants of credit spread changes. *The Journal of Finance* 56, 2177–2207. URL: <http://dx.doi.org/10.1111/0022-1082.00402>, doi:10.1111/0022-1082.00402.

- David, A., 2007. Inflation uncertainty, asset valuations, and the credit spreads puzzle. *Review of Financial Studies* 21, 2487–2534. URL: <http://dx.doi.org/10.1093/rfs/hhm041>, doi:10.1093/rfs/hhm041.
- David, A., Veronesi, P., 2013. What ties return volatilities to price valuations and fundamentals? *Journal of Political Economy* 121, 682–746. URL: <http://dx.doi.org/10.1086/671799>, doi:10.1086/671799.
- Diamond, D.W., 1991. Debt maturity structure and liquidity risk. *Quarterly Journal of Economics* 106, 709–737. doi:10.2307/2937924.
- Dick-Nielsen, J., Poulsen, T.K., 2019. How to clean academic TRACE data. Working Paper URL: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3456082.
- Diercks, A.M., Campbell, C., Sharpe, S.A., Soques, D., 2023. The swaps strike back: Evaluating expectations of one-year inflation. *Finance and Economics Discussion Series* , 1–43 URL: <http://dx.doi.org/10.17016/s.2023.061>, doi:10.17016/s.2023.061.
- Dovern, J., Fritsche, U., Slacalek, J., 2012. Disagreement among forecasters in G7 countries. *The Review of Economics and Statistics* 94, 1081–1096. URL: <https://EconPapers.repec.org/RePEc:trp:restat:v:94:y:2012:i:4:p:1081-1096>.
- Du, D., Elkamhi, R., Ericsson, J., 2019. Time-varying asset volatility and the credit spread puzzle. *The Journal of Finance* 74, 1841–1885. URL: <http://dx.doi.org/10.1111/jofi.12765>, doi:10.1111/jofi.12765.
- D’Amico, S., Kim, D.H., Wei, M., 2018. Tips from TIPS: The informational content of treasury inflation-protected security prices. *Journal of Financial and Quantitative Analysis* 53, 395–436. URL: <http://dx.doi.org/10.1017/s0022109017000916>, doi:10.1017/s0022109017000916.

- Edwards, A.K., Harris, L.E., Piwowar, M.S., 2007. Corporate bond market transaction costs and transparency. *The Journal of Finance* 62, 1421–1451. URL: <http://dx.doi.org/10.1111/j.1540-6261.2007.01240.x>, doi:10.1111/j.1540-6261.2007.01240.x.
- Eisfeldt, A.L., Herskovic, B., Liu, S., 2024. Interdealer price dispersion and intermediary capacity. NBER Working Paper, 32998, National Bureau of Economic Research URL: https://www.nber.org/system/files/working_papers/w32998/w32998.pdf.
- Eraker, B., Shaliastovich, I., Wang, W., 2015. Durable goods, inflation risk, and equilibrium asset prices. *The Review of Financial Studies* 29, 193–231. URL: <https://doi.org/10.1093/rfs/hhv049>, doi:10.1093/rfs/hhv049, arXiv:<https://academic.oup.com/rfs/article-pdf/29/1/193/24450725/hhv049.pdf>.
- Fama, E., 1981. Stock returns, real activity, inflation, and money. *American Economic Review* 71, 545–565. URL: <https://www.jstor.org/stable/1806180>.
- Fang, X., Liu, Y., Roussanov, N., 2025. Getting to the core: Inflation risks within and across asset classes. *The Review of Financial Studies* URL: <https://doi.org/10.1093/rfs/hhaf050>, doi:10.1093/rfs/hhaf050.
- Fischer, G., 2016. Investment choice and inflation uncertainty. London School of Economics and Political Science, Working Paper URL: <http://eprints.lse.ac.uk/54259/>.
- Fleckenstein, M., Longstaff, F., Lustig, H., 2016. Inflation-adjusted bonds and the inflation risk premium , 41–52 URL: <https://onlinelibrary.wiley.com/doi/abs/10.1002/9781118709207.ch3>, doi:<https://doi.org/10.1002/9781118709207.ch3>, arXiv:<https://onlinelibrary.wiley.com/doi/pdf/10.1002/9781118709207.ch3>.
- Fleckenstein, M., Longstaff, F.A., Lustig, H., 2014. The TIPS-treasury bond puzzle. *The Journal of Finance* 69, 2151–2197. URL: <http://dx.doi.org/10.1111/jofi.12032>, doi:10.1111/jofi.12032.

- Fleckenstein, M., Longstaff, F.A., Lustig, H., 2017. Deflation risk. *The Review of Financial Studies* 30, 2719–2760. URL: <http://dx.doi.org/10.1093/rfs/hhx021>, doi:10.1093/rfs/hhx021.
- Fleming, M.J., Sporn, J., 2013. Trading activity and price transparency in the inflation swap market. *Economic Policy Review* 19, 1–20. URL: <https://www.newyorkfed.org/medialibrary/media/research/epr/2013/0513flem.pdf>.
- Friewald, N., Nagler, F., 2019. Over-the-counter market frictions and yield spread changes. *The Journal of Finance* 74, 3217–3257. URL: <http://dx.doi.org/10.1111/jofi.12827>, doi:10.1111/jofi.12827.
- Friewald, N., Nagler, F., Wagner, C., 2022. Debt refinancing and equity returns. *The Journal of Finance* 77, 2287–2329. URL: <http://dx.doi.org/10.1111/jofi.13162>, doi:10.1111/jofi.13162.
- Goldstein, R., Ju, N., Leland, H., 2001. An ebit-based model of dynamic capital structure. *Journal of Business* 74, 483–512. doi:10.1086/209679.
- Gomes, J.a., Jermann, U., Schmid, L., 2016. Sticky leverage. *American Economic Review* 106, 3800–3828. URL: <http://dx.doi.org/10.1257/aer.20130952>, doi:10.1257/aer.20130952.
- Gomes, J.a., Schmid, L., 2021. Equilibrium asset pricing with leverage and default. *The Journal of Finance* 76, 977–1018. URL: <http://dx.doi.org/10.1111/jofi.12987>, doi:10.1111/jofi.12987.
- Grishchenko, O.V., Vanden, J.M., Zhang, J., 2016. The informational content of the embedded deflation option in TIPS. *Journal of Banking & Finance* 65, 1–26. URL: <http://dx.doi.org/10.1016/j.jbankfin.2015.12.004>, doi:10.1016/j.jbankfin.2015.12.004.

- Gürkaynak, R.S., Sack, B., Wright, J.H., 2007. The U.S. treasury yield curve: 1961 to the present. *Journal of Monetary Economics* 54, 2291–2304. URL: <http://dx.doi.org/10.1016/j.jmoneco.2007.06.029>, doi:10.1016/j.jmoneco.2007.06.029.
- Gürkaynak, R.S., Sack, B., Wright, J.H., 2010. The TIPS yield curve and inflation compensation. *American Economic Journal: Macroeconomics* 2, 70–92. URL: <http://dx.doi.org/10.1257/mac.2.1.70>, doi:10.1257/mac.2.1.70.
- Haubrich, J., Pennacchi, G., Ritchken, P., 2012. Inflation expectations, real rates, and risk premia: Evidence from inflation swaps. *Review of Financial Studies* 25, 1588–1629. URL: <http://dx.doi.org/10.1093/rfs/hhs003>, doi:10.1093/rfs/hhs003.
- He, Z., Khorrami, P., Song, Z., 2022. Commonality in credit spread changes: Dealer inventory and intermediary distress. *The Review of Financial Studies* 35, 4630–4673. URL: <http://dx.doi.org/10.1093/rfs/hhac004>, doi:10.1093/rfs/hhac004.
- Husted, L., Rogers, J., Sun, B., 2020. Monetary policy uncertainty. *Journal of Monetary Economics* 115, 20–36. URL: <https://www.sciencedirect.com/science/article/pii/S0304393218301661>, doi:<https://doi.org/10.1016/j.jmoneco.2019.07.009>.
- Kang, J., Pflueger, C.E., 2015. Inflation risk in corporate bonds. *The Journal of Finance* 70, 115–162. URL: <https://onlinelibrary.wiley.com/doi/10.1111/jofi.12187>, doi:10.1111/jofi.12187.
- Leland, H.E., 1998. Agency costs, risk management, and capital structure. *The Journal of Finance* 53, 1213–1243. URL: <http://dx.doi.org/10.1111/0022-1082.00051>, doi:10.1111/0022-1082.00051.
- Lu, X., Nozawa, Y., Song, Z., 2025. Inflation, default, and corporate bond returns. Johns Hopkins Carey Business School Research Paper URL: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5349911, doi:10.2139/ssrn.5349911.

- Nelson, C.R., Siegel, A.F., 1987. Parsimonious modeling of yield curves. *Journal of Business* 60, 473–489. doi:[10.1086/296409](https://doi.org/10.1086/296409).
- Reis, R., 2020. The People versus the Markets: A Parsimonious Model of Inflation Expectations. Discussion Papers 2033. Centre for Macroeconomics (CFM). URL: <https://ideas.repec.org/p/cfm/wpaper/2033.html>.
- Sack, B., Elsasser, R., 2002. Treasury inflation-indexed debt: A review of the U.S. experience. Finance and Economics Discussion Series 2002.0, 1–40. URL: <http://dx.doi.org/10.17016/feds.2002.32>, doi:[10.17016/feds.2002.32](https://doi.org/10.17016/feds.2002.32).
- Svensson, L.E.O., 1994. Estimating and interpreting forward interest rates: Sweden 1992-1994. URL: <https://www.elibrary.imf.org/view/journals/001/1994/114/001.1994.issue-114-en.xml>, doi:[10.5089/9781451853759.001](https://doi.org/10.5089/9781451853759.001).
- Weber, M., 2014. Nominal rigidities and asset pricing. Working Paper URL: https://red-files-public.s3.amazonaws.com/meetpapers/2014/paper_53.pdf.

Appendix

In this appendix, I introduce a structural model of default with debt rollover, stochastic inflation, sticky leverage, and incomplete nominal pass-through, and study its implications for yield spreads. Firms issue nominal debt while cash flows are driven by real activity and inflation. The mismatch between nominal liabilities and imperfectly adjusting assets links inflation risk to default risk. The model rationalizes the empirical findings that yield spreads decrease with expected inflation, increase with inflation uncertainty, and depend on the correlation between inflation and real asset innovations. The model also generates heterogeneity: sensitivity to inflation risk increases with default risk and decreases with refinancing intensity, while greater nominal pass-through amplifies sensitivity to inflation volatility.

Inflation Risk. Let P_t denote the price level and A_t^r real firm value. Nominal firm value is given by

$$A_t^n = A_t^r P_t^\phi, \quad (2)$$

where $\phi \in (0, 1]$ captures the degree of pass-through of inflation into firm value. When $\phi < 1$, assets are sticky and adjust only partially to changes in the price level.

Inflation evolves under the physical measure $\mathbb{P}$ as

$$\frac{dP_t}{P_t} = \mu_P^{\mathbb{P}} dt + \sigma_P dW_t^{P, \mathbb{P}}, \quad (3)$$

where $\mu_P^{\mathbb{P}}$ denotes physical expected inflation and σ_P denotes inflation volatility. I allow for a constant inflation risk premium through the market price of inflation risk λ_P . The change of measure from $\mathbb{P}$ to the nominal risk-neutral measure $\mathbb{Q}^n$ is

$$dW_t^{P, \mathbb{Q}^n} = dW_t^{P, \mathbb{P}} + \lambda_P dt. \quad (4)$$

Substituting this into the physical inflation process gives the risk-neutral inflation process

$$\frac{dP_t}{P_t} = \mu_P dt + \sigma_P dW_t^{P, \mathbb{Q}^n}, \quad (5)$$

where

$$\mu_P = \mu_P^{\mathbb{P}} - \sigma_P \lambda_P. \quad (6)$$

Thus, μ_P is the risk-neutral expected inflation rate. It differs from physical expected inflation because investors require compensation for bearing inflation risk.

With a constant real risk-free rate r_r , the nominal risk-free rate is

$$r_n = r_r + \mu_P = r_r + \mu_P^{\mathbb{P}} - \sigma_P \lambda_P. \quad (7)$$

Real asset growth and inflation shocks are correlated under the nominal risk-neutral measure:

$$dW_t^{A^r, \mathbb{Q}^n} dW_t^{P, \mathbb{Q}^n} = \rho_{A^r P} dt. \quad (8)$$

Applying Itô's lemma to $A_t^n = A_t^r P_t^\phi$, nominal asset volatility is

$$\sigma_{A^n}^2 = \sigma_{A^r}^2 + \phi^2 \sigma_P^2 + 2\phi \rho_{A^r P} \sigma_{A^r} \sigma_P. \quad (9)$$

Under the nominal risk-neutral measure, nominal firm value follows

$$\frac{dA_t^n}{A_t^n} = r_n dt + \sigma_{A^n} dW_t^{A^n, \mathbb{Q}^n}. \quad (10)$$

I set the asset payout rate equal to zero in the baseline model. If assets pay out at rate δ_A , the drift becomes $r_n - \delta_A$.

Expected inflation affects pricing through the risk-neutral expected inflation rate μ_p and therefore through the nominal risk-free rate r_n . The inflation risk premium affects pricing through the wedge between $\mu_P^{\mathbb{P}}$ and μ_p . Inflation volatility affects firm value through σ_{A^n} , with exposure governed by ϕ and $\rho_{A^r P}$.

Debt Structure and Yield Spreads. The firm issues perpetual debt with coupon C and continuously refinances a fraction m of face value P (Leland, 1998). Default occurs when nominal assets fall below a threshold:

$$\tau = \inf\{t \geq 0 : A_t^n \leq A_B^n\}. \quad (11)$$

The value of a unit claim paid at default is

$$P_B(A^n) = \left(\frac{A^n}{A_B^n}\right)^{-\gamma}, \quad (12)$$

where

$$\gamma = \frac{r_n - \frac{1}{2}\sigma_{A^n}^2 + \sqrt{\left(r_n - \frac{1}{2}\sigma_{A^n}^2\right)^2 + 2(r_n + m)\sigma_{A^n}^2}}{\sigma_{A^n}^2}. \quad (13)$$

Debt value is

$$D(A^n) = \frac{C + mP}{r_n + m} (1 - P_B(A^n)) + R A_B^n P_B(A^n). \quad (14)$$

Yield spreads are

$$y - r_n = \frac{C^*}{D(A^n; C^*, P^*)} - r_n. \quad (15)$$

Model Implications. Yield spreads decrease in expected inflation because higher inflation reduces the real burden of fixed nominal debt. Inflation volatility increases spreads by raising nominal asset risk. This effect depends on $\rho_{A^r P}$. When $\rho_{A^r P} > 0$, inflation shocks and real asset shocks reinforce each other in nominal asset volatility. When $\rho_{A^r P} < 0$, inflation partially hedges real asset risk.

The correlation between inflation and real asset growth also matters. With positive correlation, low real asset growth tends to coincide with low inflation. Since liabilities are fixed in nominal terms, low inflation increases the real burden of debt precisely when firm value

is weak. This strengthens the impact of inflation risk on credit spreads.

The model generates cross-sectional variation in inflation exposure. Firms with higher asset volatility are closer to default and exhibit stronger spread sensitivity to inflation risk. Lower refinancing intensity increases exposure to fixed nominal debt and amplifies the effect of expected inflation. Pass-through also matters. When ϕ is high, inflation risk is more strongly transmitted to nominal asset volatility. When ϕ is low, this effect is dampened. Consequently, firms with greater nominal pass-through are more sensitive to inflation volatility.

Overall, the model implies that inflation risk has a larger impact on firms with higher default probability, lower refinancing intensity, and stronger exposure of nominal asset values to inflation shocks.

This Internet Appendix contains supplemental material for the article "Inflation Risk and Yield Spread Changes".

Section [A](#) presents the main model solution. Section [B](#) reports a description of the inflation swap market and Section [C](#) details the procedure for seasonal adjustment of swap and TIPS rates. Section [D](#) details the construction of the TRACE to CRSP linking table. Section [E](#) presents additional figures and Section [F](#) additional tables.

- Table [IA.I](#) presents model parameters.
- Tables [IA.II](#) and [IA.III](#) report replication tables of [CDGM](#).
- Tables [IA.IV](#), [IA.V](#), and [IA.VI](#), and Figures [IA.VII](#), [IA.VIII](#), and [IA.IX](#), report and illustrate, group by group, the heterogeneity results summarized in Table [V](#) in Section [IV](#), for firm fundamentals (size, ROA, profit margin, rating), operating channels (operating leverage, markup, price pass-through), and debt structure (leverage, refinancing intensity, debt growth, floating-rate, maturity), respectively; the debt-structure figure additionally shows trading volume, which the corresponding table reports separately in Table [IA.VII](#).
- Table [IA.VIII](#) checks the price pass-through measure against industry gross margins (Panel A), sorts jointly on markup and price pass-through with the High-Minus-Low margins of that grid (Panel B), and re-estimates the pass-through contrast holding credit quality and industry fixed (Panel C).
- Table [IA.IX](#) re-estimates the fully-controlled specification on each subperiod, reporting both the coefficient on each inflation-risk proxy and the within- R^2 gain from adding the three of them.
- Table [IA.X](#) presents time-series regressions including all bonds regardless of industry, size, or type of bond.
- Table [IA.XI](#) presents time-series regressions including only bonds that trade on the last trading day of the month as in [Friewald and Nagler \(2019\)](#).

A. Model Solution

Let A_t^r and P_t denote the firm's real asset value and the price index, respectively. I define nominal asset value as

$$A_t^n = A_t^r P_t^\phi, \quad (\text{IA.1})$$

where $\phi \in (0, 1]$ captures the sensitivity of nominal assets to inflation. When $\phi = 1$, nominal assets fully adjust to inflation, whereas when $\phi < 1$, pass-through is incomplete and assets are sticky in nominal terms.

I assume that inflation follows a geometric Brownian motion under the physical measure $\mathbb{P}$:

$$\frac{dP_t}{P_t} = \mu_P^\mathbb{P} dt + \sigma_P dW_t^{P, \mathbb{P}}. \quad (\text{IA.2})$$

The market price of inflation risk is denoted by λ_P . The change of measure from $\mathbb{P}$ to the nominal risk-neutral measure $\mathbb{Q}^n$ is

$$dW_t^{P, \mathbb{Q}^n} = dW_t^{P, \mathbb{P}} + \lambda_P dt. \quad (\text{IA.3})$$

Therefore, under $\mathbb{Q}^n$,

$$\frac{dP_t}{P_t} = \mu_P dt + \sigma_P dW_t^{P, \mathbb{Q}^n}, \quad (\text{IA.4})$$

where

$$\mu_P = \mu_P^\mathbb{P} - \sigma_P \lambda_P. \quad (\text{IA.5})$$

The variable μ_P is the risk-neutral expected inflation rate. Since the debt claims in the model are nominal, the relevant pricing measure is the nominal risk-neutral measure $\mathbb{Q}^n$.

Let r_r denote the real risk-free rate. The nominal risk-free rate is

$$r_n = r_r + \mu_P. \quad (\text{IA.6})$$

Under the nominal risk-neutral measure, I write the real asset process as

$$\frac{dA_t^r}{A_t^r} = \delta_{A^r}^n dt + \sigma_{A^r} dW_t^{A^r, \mathbb{Q}^n}, \quad (\text{IA.7})$$

where the Brownian motions $dW_t^{A^r, \mathbb{Q}^n}$ and $dW_t^{P, \mathbb{Q}^n}$ have instantaneous correlation

$$dW_t^{A^r, \mathbb{Q}^n} dW_t^{P, \mathbb{Q}^n} = \rho_{A^r P} dt. \quad (\text{IA.8})$$

Since

$$A_t^n = A_t^r P_t^\phi, \quad (\text{IA.9})$$

Itô's lemma implies

$$\frac{dA_t^n}{A_t^n} = \frac{dA_t^r}{A_t^r} + \phi \frac{dP_t}{P_t} + \phi \frac{dA_t^r}{A_t^r} \frac{dP_t}{P_t} + \frac{1}{2} \phi(\phi - 1) \sigma_P^2 dt. \quad (\text{IA.10})$$

Substituting the processes for A_t^r and P_t gives

$$\frac{dA_t^n}{A_t^n} = \left[\delta_{A^r}^n + \phi\mu_p + \phi\rho_{A^rP}\sigma_P\sigma_{A^r} + \frac{1}{2}\phi(\phi-1)\sigma_P^2 \right] dt + \sigma_{A^r}dW_t^{A^r,\mathbb{Q}^n} + \phi\sigma_PdW_t^{P,\mathbb{Q}^n}. \quad (\text{IA.11})$$

Because A_t^n represents the value of traded assets in nominal units, its expected return under $\mathbb{Q}^n$ must equal the nominal risk-free rate. In the baseline model, I set the asset payout rate equal to zero. Therefore, the nominal asset process satisfies

$$\frac{dA_t^n}{A_t^n} = r_ndt + \sigma_{A^n}dW_t^{A^n,\mathbb{Q}^n}. \quad (\text{IA.12})$$

Matching the drift of the Itô expression to $r_n = r_r + \mu_p$ implies

$$\delta_{A^r}^n = r_r + (1-\phi)\mu_p - \phi\rho_{A^rP}\sigma_P\sigma_{A^r} - \frac{1}{2}\phi(\phi-1)\sigma_P^2. \quad (\text{IA.13})$$

Thus, under the nominal risk-neutral measure, the real asset process follows

$$\frac{dA_t^r}{A_t^r} = \left[r_r + (1-\phi)\mu_p - \phi\rho_{A^rP}\sigma_P\sigma_{A^r} - \frac{1}{2}\phi(\phi-1)\sigma_P^2 \right] dt + \sigma_{A^r}dW_t^{A^r,\mathbb{Q}^n}. \quad (\text{IA.14})$$

Substituting this expression back into the nominal asset dynamics gives

$$\frac{dA_t^n}{A_t^n} = r_ndt + \sigma_{A^r}dW_t^{A^r,\mathbb{Q}^n} + \phi\sigma_PdW_t^{P,\mathbb{Q}^n}. \quad (\text{IA.15})$$

Defining total nominal asset variance as

$$\sigma_{A^n}^2 = \sigma_{A^r}^2 + \phi^2\sigma_P^2 + 2\phi\rho_{A^rP}\sigma_{A^r}\sigma_P, \quad (\text{IA.16})$$

I can write the nominal asset process compactly as

$$\frac{dA_t^n}{A_t^n} = r_ndt + \sigma_{A^n}dW_t^{A^n,\mathbb{Q}^n}. \quad (\text{IA.17})$$

The firm commits to a stationary debt structure by issuing consol bonds to optimally set total debt through a constant coupon rate C . Following [Leland \(1998\)](#), the firm continuously retires a fraction mP of its outstanding debt at par, where P denotes the total face value of debt and m is the refinancing intensity, with $0 < m \leq 1$. The retired debt is immediately replaced with newly issued debt of identical maturity, coupon, principal, and seniority, which ensures a constant rollover structure.

Debt issuance provides a tax advantage $\tau_{tax}C$ but also increases bankruptcy costs borne by equity holders. Coupon payments can be funded either out of cash flow plus the tax shield or, if necessary, out of equity holders' own resources. Default occurs when equity holders deem firm value too low to justify further payments. Upon liquidation, debt holders recover the residual value of the firm net of bankruptcy costs, determined by the recovery rate R .

Equity holders are the residual claimants on the firm's cash flows. Therefore, equity value,

E , is given by the difference between levered firm value, $\mathbf{v}$, and debt value, D . If A_t^n declines to a critical endogenous threshold, A_B^n , equity holders default. Let τ denote the first time assets hit A_B^n :

$$\tau = \inf\{t \geq 0 : A_t^n \leq A_B^n\}. \quad (\text{IA.18})$$

Let $r_n = r_r + \mu_p$. The value of a unit claim at default is

$$P_B(A^n) = \left(\frac{A^n}{A_B^n}\right)^{-\gamma}, \quad (\text{IA.19})$$

$$\gamma = \frac{r_n - \frac{1}{2}\sigma_{A^n}^2 + \sqrt{\left(r_n - \frac{1}{2}\sigma_{A^n}^2\right)^2 + 2(r_n + m)\sigma_{A^n}^2}}{\sigma_{A^n}^2}.$$

Debt value is then given by

$$D(A^n; A_B^n, P, C) = \frac{C + mP}{r_n + m} [1 - P_B(A^n)] + RA_B^n P_B(A^n), \quad (\text{IA.20})$$

where the first term is the value of coupon and principal payments up to default and the second term is the recovery value in bankruptcy.

The total value of the firm, $\mathbf{v}$, equals asset value plus tax benefits minus bankruptcy costs:

$$\mathbf{v}(A^n; A_B^n, P, C) = A^n + TB(A^n; A_B^n, C) - BC(A^n; A_B^n). \quad (\text{IA.21})$$

Define $x = \gamma$ when $m = 0$. Equivalently,

$$x = \frac{r_n - \frac{1}{2}\sigma_{A^n}^2 + \sqrt{\left(r_n - \frac{1}{2}\sigma_{A^n}^2\right)^2 + 2r_n\sigma_{A^n}^2}}{\sigma_{A^n}^2}. \quad (\text{IA.22})$$

Then the present value of bankruptcy costs is

$$BC(A^n; A_B^n) = (1 - R)A_B^n \left(\frac{A^n}{A_B^n}\right)^{-x}. \quad (\text{IA.23})$$

The value of the coupon stream up to default is

$$\frac{C}{r_n} \left[1 - \left(\frac{A^n}{A_B^n}\right)^{-x} \right], \quad (\text{IA.24})$$

which implies a tax benefit of debt equal to

$$TB(A^n; A_B^n, C) = \frac{\tau_{tax}C}{r_n} \left[1 - \left(\frac{A^n}{A_B^n}\right)^{-x} \right]. \quad (\text{IA.25})$$

Since the firm has only equity and debt outstanding, firm value satisfies

$$\begin{aligned} v(A^n; A_B^n, P, C) &= E(A^n; A_B^n, P, C) + D(A^n; A_B^n, P, C) \\ &= A^n + TB(A^n; A_B^n, C) - BC(A^n; A_B^n). \end{aligned} \quad (\text{IA.26})$$

Since $E = v - D$, equity value is

$$\begin{aligned} E(A^n; A_B^n, P, C) &= A^n + \frac{\tau_{tax}C}{r_n} \left[1 - \left(\frac{A^n}{A_B^n} \right)^{-x} \right] - (1-R)A_B^n \left(\frac{A^n}{A_B^n} \right)^{-x} \\ &\quad - \frac{C+mP}{r_n+m} [1 - P_B(A^n)] - RA_B^n P_B(A^n). \end{aligned} \quad (\text{IA.27})$$

Since equity is a limited-liability claim, equity holders never allow equity value to become negative. Instead, they stop servicing debt and force the firm into bankruptcy. Therefore, the optimal default-triggering level satisfies the smooth-pasting condition

$$\left. \frac{d}{dA^n} E(A^n; A_B^n, P, C) \right|_{A^n=A_B^n} = 0. \quad (\text{IA.28})$$

This condition yields

$$A_B^n = \frac{\frac{\gamma(C+mP)}{r_n+m} - \frac{\tau_{tax}Cx}{r_n}}{1 + (1-R)x + R\gamma}. \quad (\text{IA.29})$$

Plugging this solution into equation (IA.20) gives a closed-form expression for total debt value, conditional on coupon C and principal value P . When debt is initially issued, there is an additional restriction linking market value, coupon, and principal: the coupon is set such that market debt value, D , equals principal value, P . That is, debt is issued at par. This requires C to solve

$$D(A^n; A_B^n, P, C) = P. \quad (\text{IA.30})$$

At time $t = 0$, the initial owners choose the optimal debt policy to maximize total firm value. Equivalently, for each candidate principal P , the coupon is chosen to satisfy the par-pricing condition, and the firm selects the debt policy that maximizes firm value:

$$(C^*, P^*) = \arg \max_{C, P} v(A_0^n; A_B^n(C, P), P, C), \quad (\text{IA.31})$$

subject to

$$D(A_0^n; A_B^n(C, P), P, C) = P. \quad (\text{IA.32})$$

Yield spreads are therefore given by

$$y - r_n = \frac{C^*}{D(A_0^n; A_B^n(C^*, P^*), P^*, C^*)} - r_n. \quad (\text{IA.33})$$

When debt is issued at par, this simplifies to

$$y - r_n = \frac{C^*}{P^*} - r_n. \quad (\text{IA.34})$$

A.1. Model Implications

To illustrate how yield spreads relate to inflation risk, I simulate the model 1000 times by drawing parameter values for expected inflation, μ_p , inflation volatility, σ_p , and the correlation between inflation and asset growth, $\rho_{A'P}$, from reasonable parameter intervals. Throughout the simulations, I set the inflation risk premium to zero, $\lambda_p = 0$, so physical and risk-neutral expected inflation coincide. Expected inflation ranges from -2% to 10% , inflation volatility from 0.5% to 10% , and the correlation between inflation and assets from -1 to 1 . All other parameters are fixed at values commonly used in the literature, as summarized in Table IA.I.³⁰

Table IA.I
about
here

A.1.1. Implications for Yield Spreads

Figure IA.I illustrates the relationship between inflation risk and yield spreads, together with the corresponding regression lines. Blue dots denote observations for which the inflation-asset growth correlation, $\rho_{A'P}$, is positive, while red dots denote observations for which it is negative.

Figure IA.I
about
here

Panel A shows that yield spreads decrease with expected inflation. An increase in expected inflation lowers the real value of nominal debt, since debt is denominated in nominal terms. This reduction in the real burden of debt lowers default risk and, in turn, narrows yield spreads.

Panel B shows the effect of inflation volatility on yield spreads. Higher inflation volatility increases the volatility of nominal assets, thereby raising default risk and widening spreads. The strength of this effect depends on the correlation between inflation and real asset growth. When asset-inflation correlation is positive, the direct effect of inflation volatility and the covariance term in nominal asset volatility reinforce each other, generating a stronger positive relationship between inflation volatility and yield spreads.³¹ When correlation is negative, the covariance term offsets part of the direct volatility effect, making the relationship between yield spreads and inflation volatility flatter. The dispersion in Figure IA.I, Panel B, reflects this asymmetric response of nominal asset volatility to inflation volatility depending on the sign of the asset-inflation correlation.

Panel C highlights the role of the correlation between inflation and real asset innovations, which captures the cyclicity of inflation risk. When this correlation is positive, inflation shocks and real asset shocks reinforce each other in nominal asset volatility. This increases

³⁰The pass-through parameter ϕ is set to 0.4 as in Bhamra, Dorion, Jeanneret and Weber (2022), while the recovery rate R and tax rate τ_{ax} are set to 0.5 and 0.35, respectively, as in Leland (1998) and Du, Elkamhi and Ericsson (2019).

³¹Since inflation-growth correlation is unconditionally positive in the sample, the amplification mechanism under positive $\rho_{A'P}$ is the empirically most relevant case.

default risk and widens yield spreads. When the correlation is negative, inflation shocks partially hedge real asset risk, dampening the effect of inflation volatility on spreads.

A.1.2. *Heterogeneity Implications*

The model also generates predictions about heterogeneity, particularly with respect to asset volatility, the degree of nominal pass-through, ϕ , and refinancing intensity. More generally, it suggests that the effect of inflation risk on a firm depends on both its distance to default and its exposure to inflation shocks.

Figure IA.II shows that, holding nominal debt constant, higher asset volatility reduces firm value and raises measured leverage. This moves firms closer to their default boundary and amplifies the sensitivity of spreads to asset risk. As a result, firms with higher leverage experience stronger effects from both expected inflation and inflation volatility.

Figure IA.II
about
here

In addition, as in the model of Friewald, Nagler and Wagner (2022), leverage and refinancing intensity have opposite implications for default probabilities, with lower refinancing intensity being associated with higher default risk. Thus, for a given leverage level, lower refinancing intensity increases the sensitivity of spreads to inflation risk, as shown in Figure IA.III.

Figure IA.II
about
here

Figure IA.IV shows that cash-flow or asset stickiness determines the extent to which inflation risk passes through to nominal asset volatility. As shown in the expression for σ_{A^n} , when assets are more flexible, so that ϕ is closer to one, inflation shocks have a larger effect on nominal asset risk. When assets are stickier, so that $\phi < 1$, this effect is dampened. Consequently, firms with greater nominal pass-through are more sensitive to inflation volatility and to the interaction between inflation and real asset growth than firms whose nominal cash flows adjust less fully.

Figure IA.IV
about
here

The price pass-through sort in Section IV is the empirical counterpart of the model's pass-through parameter ϕ . A higher ϕ means that a firm's nominal asset value moves more closely with the price level. Empirically, I proxy for this exposure using the extent to which an industry's output prices move relative to the input-weighted prices of its suppliers. Consistent with this interpretation, Table IA.VIII shows that gross margins rise with own-industry producer-price inflation in high-pass-through industries but are flat in the low and middle groups, although the difference between the extreme groups is not statistically significant. I therefore interpret the highest pass-through group as corresponding to firms with high ϕ .

In the model, once nominal drift is pinned down by no-arbitrage, ϕ affects yield spreads mainly through nominal asset volatility. The clearest prediction is therefore that inflation uncertainty should matter more when pass-through is high. Figure IA.IV confirms this pattern in the model, and the empirical estimates point in the same direction, with substantially stronger sensitivity to inflation uncertainty among high-pass-through firms.

The expected-inflation results also become stronger with pass-through, but much more than the model alone predicts. This points to an additional operating channel that the model holds fixed. When firms can pass higher costs into output prices, inflation raises nominal revenues as well as nominal asset values, increasing the sensitivity of credit risk to expected inflation. The markup results are consistent with the same mechanism, as expected-inflation sensitivity is stronger for low-markup firms, whose prices lie closer to costs.

Although stylized, the model captures the key channels through which inflation risk affects

yield spreads. In particular, it rationalizes yield spreads decreasing with expected inflation and increasing with inflation volatility and cyclicalities. Consistent with the empirical findings, it further implies that inflation risk has a larger impact on firms with higher default probability and lower refinancing intensity, while greater nominal pass-through amplifies sensitivity to inflation volatility.

B. Inflation Swaps and Market-Based Inflation Expectations

In 1997, the U.S. Treasury started issuing Treasury Inflation-Protected Securities, fixed-coupon bonds whose principal amount is adjusted daily based on the Consumer Price Index (CPI) for All Urban Consumers in the third preceding calendar month. From the first TIPS auction onward, market participants made markets in inflation derivatives as a way of hedging inflation risk. Zero-coupon inflation swaps quickly became one of the most liquid inflation derivatives in the over-the-counter market. These swaps are forward contracts in which the buyer pays a fixed nominal rate and receives an inflation-linked payment from the seller. They are quoted with maturities ranging from 1 to 30 years and, along with nominal Treasuries, offer an alternative means of measuring real yields.

I use inflation swap rates as the main market-based proxy for inflation expectations for several reasons. First, inflation swaps tend to predict future inflation rates more accurately than surveys. As shown by [Diercks, Campbell, Sharpe and Soques \(2023\)](#), inflation swaps align more closely with realized inflation rates than survey forecasts, which exhibit less variation. Although inflation swaps display higher variance than survey expectations, they remain less volatile than realized inflation. In [Figure IA.V](#), consistent with [Diercks, Campbell, Sharpe and Soques \(2023\)](#), I show that surveys tend to cluster most forecasts around 2% compared to the actual realized inflation, while inflation swap rates generally show more variation.

Second, at medium to long horizons, the market for inflation protection is deep and responsive to fundamentals, although swap rates still contain an inflation risk premium. [Bahaj, Czech, Ding and Reis \(2023\)](#) utilize transaction-level data from UK inflation swaps and show that the supply of long-term inflation protection is highly elastic, reflects economic fundamentals, and rapidly adjusts to new information. Finally, compared to TIPS, inflation swaps are less affected by the embedded deflation option and by liquidity effects, for several reasons. The TIPS inflation adjustment is bounded below at its issuance value providing an embedded put option that protects investors against deflation on the bond's principal payment (e.g., [Grishchenko, Vanden and Zhang \(2016\)](#); [Christensen, Lopez and Rudebusch \(2015\)](#)). Because this option has a nonnegative value, it lowers TIPS yields compared to bonds that are fully indexed to inflation. Zero-coupon inflation swap contracts do not contain this option. Therefore, all else equal, the break-even inflation rate (Treasury rate minus TIPS rate) based on a TIPS principal strip should be higher than the equivalent maturity inflation swap rate. In addition to the deflation option, studies by [Sack and Elsasser \(2002\)](#), [Fleckenstein, Longstaff and Lustig \(2014\)](#), [D'Amico, Kim and Wei \(2018\)](#), and [Andreasen, Christensen and Riddell \(2021\)](#) consistently show that TIPS break-even inflation rates fall below survey-based inflation expectations and that Treasury bonds are almost always overvalued relative to inflation-swapped TIPS. This mispricing narrows as additional capital flows into the markets and as liquidity increases.

Figure [IA.V](#)
about
here

Figure [IA.V](#)
about
here

Even in the most recent sample, the pattern is consistent with previous studies. Figure IA.VI reports the inflation swap rate minus TIPS implied break-even rate and shows that it exhibits time variation, with a positive average and peaking during periods of low liquidity. This evidence is consistent with Campbell, Shiller and Viceira (2012) and Haubrich, Pennacchi and Ritchken (2012), who attribute the spike in TIPS yields after Lehman Brothers’ bankruptcy to Lehman’s extensive use of TIPS for collateralizing its repo borrowings and derivative positions and with Fleckenstein, Longstaff and Lustig (2014) which finds that the price difference narrows when the U.S. auctions nominal Treasuries or TIPS, and it widens when dealers have difficulty obtaining Treasury securities, such as during a period of increased repo failures. Therefore, TIPS yields contain a liquidity premium because, like other bonds, they are held in buy-and-hold investors’ portfolios, causing break-even inflation rates to diverge further from inflation swap rates.

Overall, these factors motivate the use of inflation swaps as the baseline market-based proxy, with the survey-anchored curve of Section III.C as the complementary check on their risk-premium and liquidity components.

C. Seasonal Adjustment of Swap and TIPS rates

TIPS and zero-coupon inflation swaps are both indexed to the non-seasonally adjusted U.S. CPI index, thus seasonal patterns in inflation must be taken into account when matching with corporate bond cash flows for swap maturities that include fractional years (e.g., 7.5 years). I adjust swap and TIPS rates following the procedure of Fleckenstein, Longstaff and Lustig (2014). Specifically, I fit a standard cubic spline through the quoted maturities of both swaps and TIPS using a grid size of one month. I then estimate seasonal components in inflation from the monthly non-seasonally adjusted U.S. CPI index (CPI-U NSA) series between January 1980 and December 2021 by estimating a regression of monthly log changes in the CPI index on month dummies. I obtain an estimate of the seasonal effect in each month. I normalize these seasonal factors so that their product is unity, thus, by construction, there will be no seasonal adjustment for full-year maturities. Next, monthly forward rates are constructed from the interpolated rates, and multiplied by the corresponding adjustment factor. Lastly, I obtain seasonally adjusted rates by converting forward rates into spot rates. As suggested by Fleckenstein, Longstaff and Lustig (2014), since the interpolated rates would then be sensitive to short-term inflation assumptions, I do not interpolate or adjust maturities shorter than the quoted ones, i.e., one year for the swaps and two years for the TIPS, but instead use the shortest quoted maturity rate.

D. Linking Academic TRACE to CRSP

I create a linking table to match TRACE CUSIPs to CRSP permco, accounting for firm mergers, delistings, and splits. First, I merge TRACE with Mergent FISD to obtain issuer and parent CUSIPs. For the CRSP merge, I rely on 6-digit CUSIPs and the TRACE-reported trading symbol, which corresponds to a firm’s ticker.

I begin by matching CRSP to TRACE using issuer CUSIPs, tracking the issuing firm forward through the CRSP delisting table to account for mergers or spin-offs. Trading

symbols help resolve cases where a bond corresponds to multiple firms on the same date, prioritizing matches where the TRACE trading symbol aligns with the CRSP company symbol.

For unmatched bonds, I match them to CRSP firms using parent CUSIPs. By tracing the parent firm back in time to the bond's offering date and utilizing trading symbols, I identify the correct firm path. First, I ensure the path reaches the first symbol match before other firms, and if multiple firm-date matches occur, I prioritize matches where the TRACE trading symbol aligns with the CRSP company symbol at the time. If no valid match exists, I use the symbol match at any point, provided no alternative matches exist.

Next, I match firms using trading symbols, requiring that symbols are active simultaneously in both CRSP and TRACE databases. Matches are valid only when dates overlap, and the longest period of valid symbol matches is retained. After removing six instances of multiple firm-bond matches, I match the remaining bonds using company names, first matching the issuing firm from FISD with CRSP at the offering date, and then matching parent firm names for any remaining bonds.

E. Additional Figures

Figure IA.I: Model Simulation

This figure shows the yield spread implications of inflation risk. I simulate the model 1000 times where I fix all parameters except the expected inflation rate (μ_P), inflation volatility (σ_P), and the correlation between inflation and assets (ρ_{A^*P}). I uniformly draw these parameters from reasonable intervals: expected inflation varies between -2% and 10%, inflation volatility between 0.5% and 10%, and correlation between inflation and real assets between -1 and 1. Table IA.I provides the details about all parameter choices. I also report regression lines. In Panel (b), the regression line is conditional on the correlation between inflation and real assets.

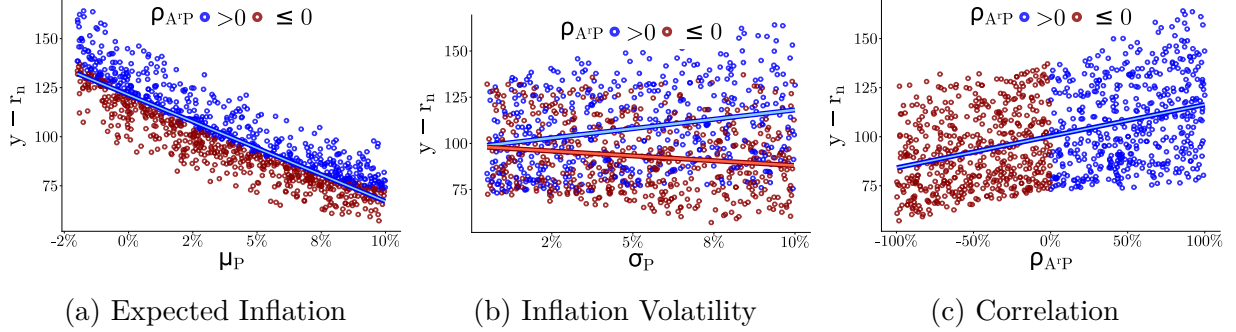


Figure IA.II: Heterogeneity - Asset Volatility

This figure shows the heterogeneity implications of inflation risk. I calibrate the model for each level of real asset volatility from 5 to 50% and for each level of expected inflation rate (μ_P), inflation volatility (σ_P), and the correlation between inflation and assets (ρ_{A^*P}). Expected inflation varies between -2% and 10%, inflation volatility between 0.5% and 10%, and correlation between inflation and real assets between -1 and 1. I fix all parameters according to Table IA.I.

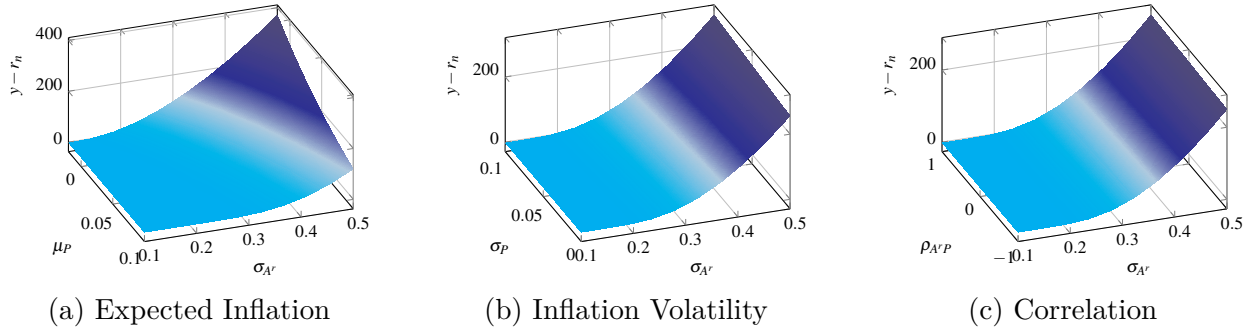


Figure IA.III: **Heterogeneity - Refinancing Intensity**

This figure shows the heterogeneity implications of inflation risk. I calibrate the model for each level of refinancing intensity (m) from 0 to 1 and for each level of expected inflation rate (μ_P), inflation volatility (σ_P), and the correlation between inflation and assets (ρ_{AP}). Expected inflation varies between -2% and 10%, inflation volatility between 0.5% and 10%, and correlation between inflation and real assets between -1 and 1. I fix all parameters according to Table IA.I.

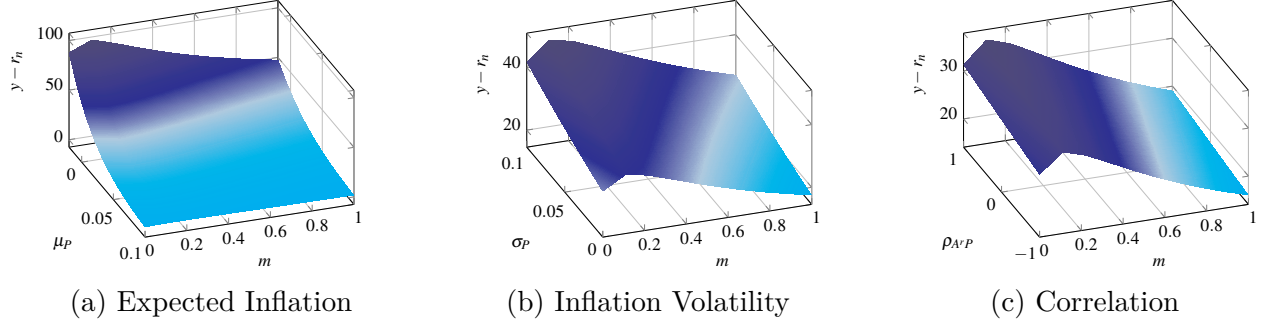


Figure IA.IV: **Heterogeneity - Nominal Pass-Through**

This figure shows the heterogeneity implications of inflation risk. I calibrate the model for each level of the pass-through parameter (ϕ) from 0.1 to 0.9 and for each level of expected inflation rate (μ_P), inflation volatility (σ_P), and the correlation between inflation and assets (ρ_{AP}). Expected inflation varies between -2% and 10%, inflation volatility between 0.5% and 10%, and correlation between inflation and real assets between -1 and 1. I fix all parameters according to Table IA.I.

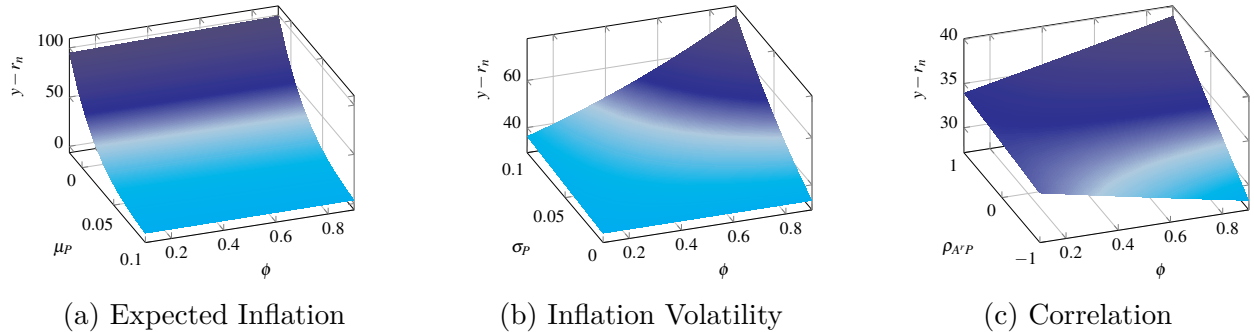


Figure IA.V: **Histogram of Forecasts vs Realized: Surveys and Swaps**

This figure reports the histogram of the forecasts and the 1-year inflation rates. The horizontal axis reflects the forecast or realized inflation in percentage points, while the vertical axis captures the frequency. In blue I report the realizations while in red the expectations. Panel (a) shows the expected price change for one year from Consumer Surveys at the University of Michigan, while Panel (b) shows the one-year inflation swaps.

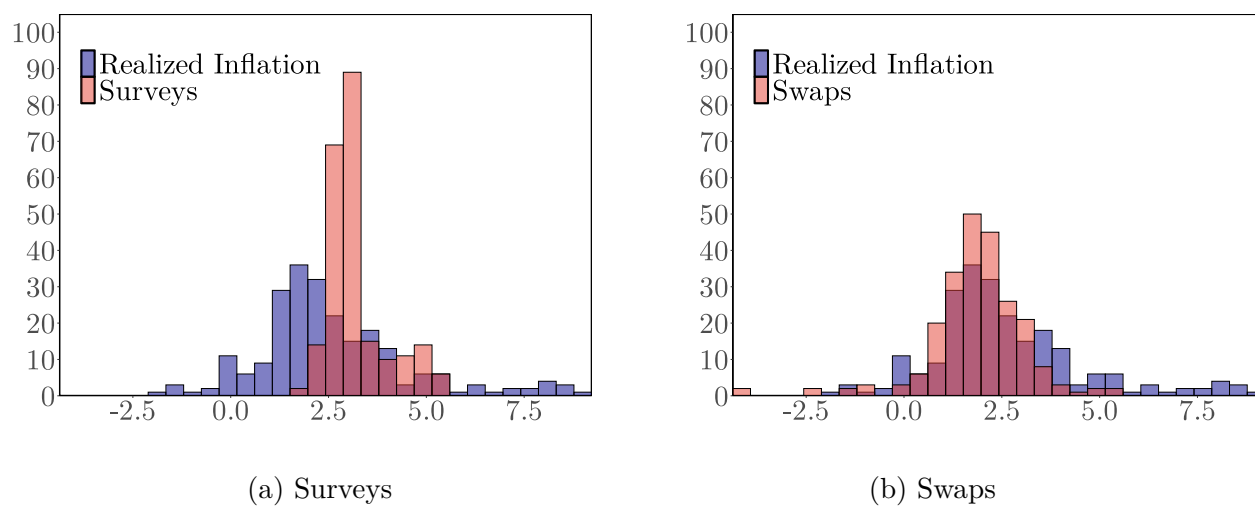


Figure IA.VI: Time Series of Treasury, TIPS and Swap rates

The top panel shows the time series of 2-year zero coupon treasury yield, break-even, and inflation swaps. The bottom panel represents the difference between the 2-year zero coupon inflation swap rate and the 2-year TIPS implied zero-coupon break-even inflation yield. Yields are expressed as annual percentages, and the difference is in annual basis points.

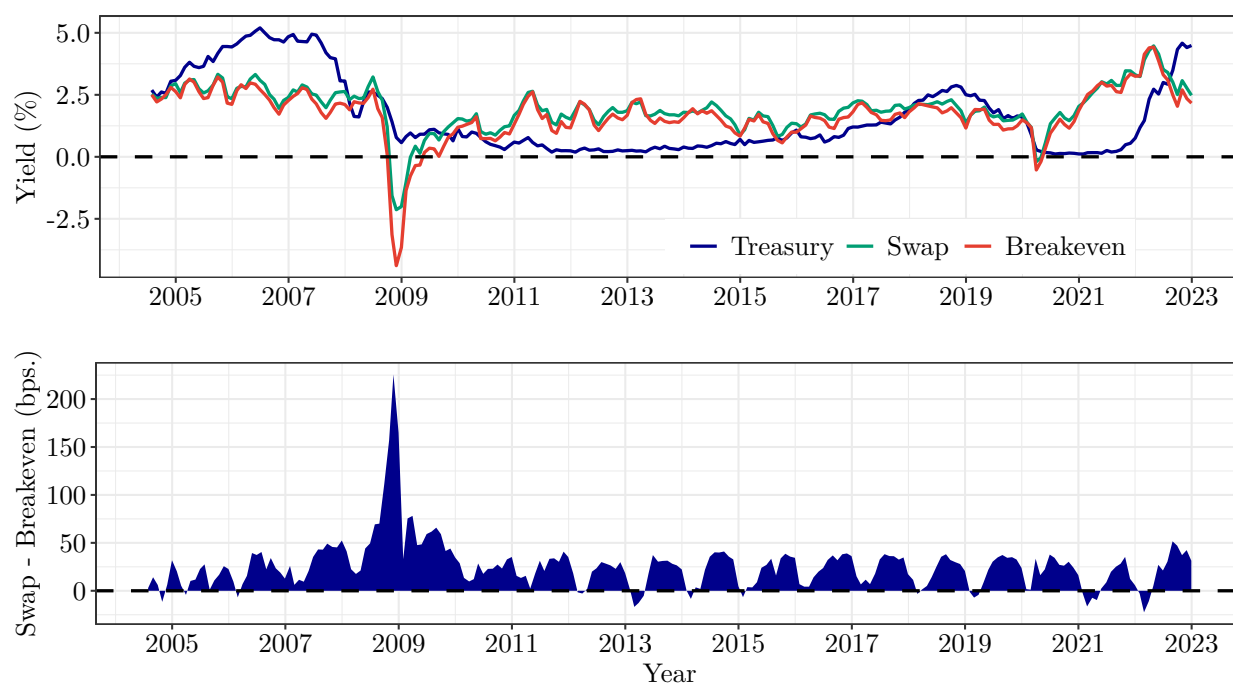


Figure IA.VII: **Heterogeneity in Inflation Risk: Firm Fundamentals**

For each heterogeneity dimension, I estimate the bond-fixed-effects panel regression

$$\Delta YS_{i,t} = \alpha_i + \beta^T \Delta S_{i,t} + \Gamma^T \Delta C_{i,t} + \sum_g (\theta_g^T \Delta I_{i,t}) \mathbb{1}\{G_{i,t} = g\} + v_{i,t},$$

using industrial bonds with at least 25 monthly observations of yield spread changes $\Delta YS_{i,t}$. Here, $\Delta S_{i,t}$ represents the [CDGM](#) structural credit-risk variables, $\Delta C_{i,t}$ contains the intermediation controls from [Friewald and Nagler \(2019\)](#), [He, Khorrami and Song \(2022\)](#), and [Eisfeldt, Herskovic and Liu \(2024\)](#), and $\Delta I_{i,t}$ denotes the inflation-risk proxies ($\Delta E[\mu^S]_{i,t}$, $\Delta \sigma_{i,t}^S$, $\Delta \rho_t^S$) introduced in [Section III](#). The coefficients on the credit-risk and intermediation controls are common across groups, while the inflation-risk coefficients θ_g vary with the sorting characteristic. Groups are formed using the value of the relevant characteristic one month before the observation, except for size, which is measured at issuance. The figure reports the group-specific inflation-risk coefficients and 95% confidence intervals based on standard errors two-way clustered by bond and month. Columns correspond to size, return on assets, profit margin, and credit rating, while rows correspond to expected inflation, inflation uncertainty, and inflation cyclicalities. The sample is based on U.S. industrial corporate bond transaction data from TRACE for the period 2004–2021.

Figure IA.VIII: **Heterogeneity in Inflation Risk: Operating Channels**

For each industrial bond i with at least 25 monthly observations of yield spread changes $\Delta YS_{i,t}$, I estimate one bond-fixed-effects panel regression per heterogeneity dimension,

$$\Delta YS_{i,t} = \alpha_i + \beta^T \Delta S_{i,t} + \Gamma^T \Delta C_{i,t} + \sum_g (\theta_g^T \Delta I_{i,t}) \mathbb{1}\{G_{i,t} = g\} + v_{i,t},$$

where $\Delta S_{i,t}$ contains the [CDGM](#) structural credit-risk variables, $\Delta C_{i,t}$ the intermediation controls from [FN](#), [HKS](#), and [EHL](#), and $\Delta I_{i,t}$ the inflation-risk proxies ($\Delta E[\mu^S]_{i,t}$, $\Delta \sigma_{i,t}^S$, $\Delta \rho_t^S$) introduced in [Section III](#). The control coefficients are common across groups, while the inflation-risk coefficients vary with the sorting characteristic. Operating leverage and markup are measured one month before the observation, while price pass-through uses the expanding mean through twelve months before the observation. The figure reports the group-specific inflation-risk coefficients and 95% confidence intervals based on standard errors two-way clustered by bond and month. Columns correspond to operating leverage, markup, and price pass-through, while rows correspond to expected inflation, inflation uncertainty, and inflation cyclicality. The sample is based

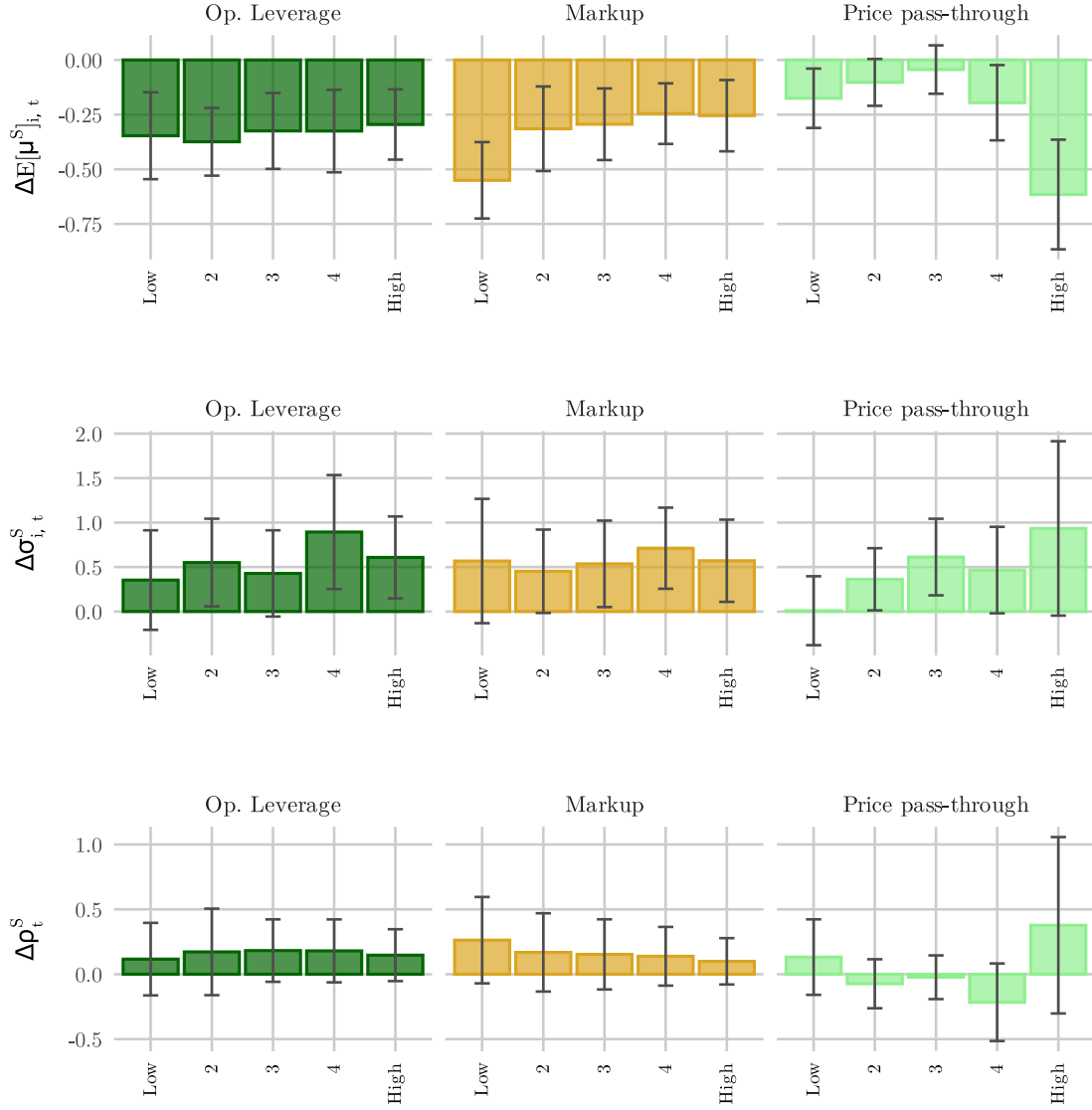
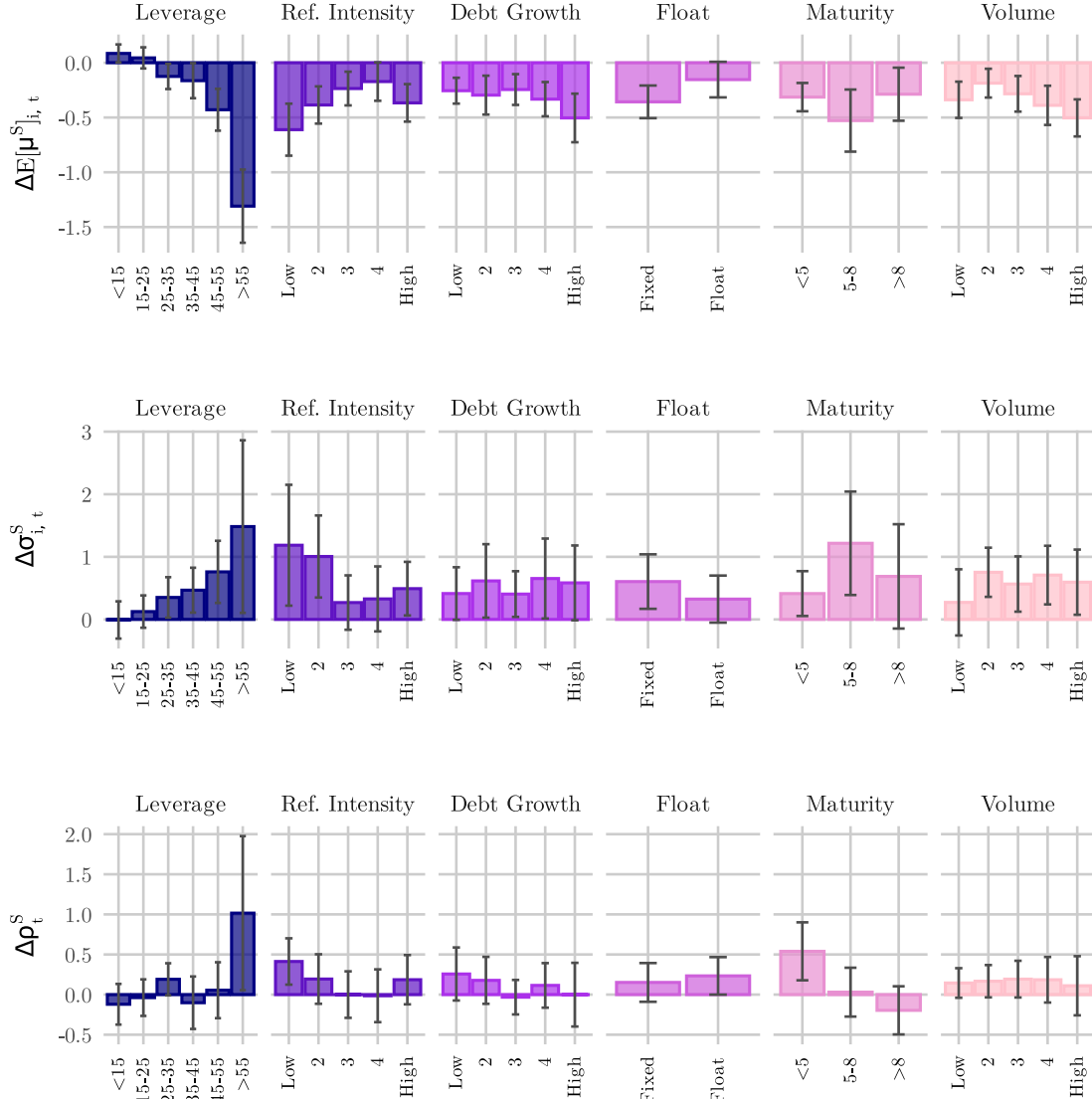


Figure IA.IX: **Heterogeneity in Inflation Risk: Debt Structure and Liquidity**

For each industrial bond i with at least 25 monthly observations of yield spread changes $\Delta YS_{i,t}$, I estimate one bond-fixed-effects panel regression per heterogeneity dimension,

$$\Delta YS_{i,t} = \alpha_i + \beta^T \Delta S_{i,t} + \Gamma^T \Delta C_{i,t} + \sum_g (\theta_g^T \Delta I_{i,t}) \mathbb{1}\{G_{i,t} = g\} + v_{i,t},$$

where $\Delta S_{i,t}$ contains the [CDGM](#) structural credit-risk variables, $\Delta C_{i,t}$ the intermediation controls from [FN](#), [HKS](#), and [EHL](#), and $\Delta I_{i,t}$ the inflation-risk proxies ($\Delta E[\mu^S]_{i,t}$, $\Delta \sigma^S_{i,t}$, $\Delta \rho^S_t$) introduced in Section [III](#). The control coefficients are common across groups, while the inflation-risk coefficients vary with the sorting characteristic. Leverage, refinancing intensity, floating-rate exposure, maturity, and trading volume are measured one month before the observation, while debt growth uses the expanding mean of the underlying monthly series through twelve months before the observation. The figure reports the group-specific inflation-risk coefficients and 95% confidence intervals based on standard errors two-way clustered by bond and month. Columns correspond to leverage, refinancing intensity, debt growth, floating-rate exposure, maturity, and trading volume, while rows correspond to expected inflation, inflation uncertainty, and inflation cyclicality. The sample is based on U.



F. Additional Tables

Table IA.I: **Model Parameters**

This table lists the parameters used to examine how yield spreads relate to inflation risk. I fix all parameters except for expected inflation (μ_P), inflation volatility (σ_P) and the correlation between inflation and asset growth ($\rho_{A^r,P}$). I uniformly draw these parameters in reasonable intervals.

Name	Symbol	Value
Initial Price Level	P_0	1
Initial Asset Value	A_0	100
Pass-Through Parameter	ϕ	0.40
Refinancing Intensity	m	0.125
Firm-Specific Volatility	σ_{A^r}	0.30
Real Risk-Free Rate	r_r	0.04
Tax Rate	τ_{tax}	0.35
Recovery Rate	R	0.50
Expected Inflation	μ_P	$\mathcal{U}[-0.02, 0.10]$
Correlation Inflation Asset Growth	$\rho_{A^r,P}$	$\mathcal{U}[-1, 1]$
Inflation Volatility	σ_P	$\mathcal{U}[0.005, 0.10]$

Table IA.II: Determinants of Yield Spread Changes in [Collin-Dufresne, Goldstein and Martin \(2001\)](#)

For each industrial bond i with at least 25 monthly observations of yield spread changes, $\Delta YS_{i,t}$, I estimate the model:

$$\Delta YS_{i,t} = \alpha_i + \boldsymbol{\beta}_i^T \boldsymbol{\Delta S}_{i,t} + \varepsilon_{i,t},$$

where $\boldsymbol{\Delta S}_{i,t} := [\Delta Lev_{i,t}, \Delta RF_t, \Delta RF_t^2, \Delta Slope_t, \Delta VIX_t, RM_t, \Delta Jump_t]$ is the vector of the structural model variables, with $\Delta Lev_{i,t}$ as the change in firm leverage, ΔRF_t the change in 10-year Treasury interest rate, ΔRF_t^2 the squared change in the 10-year Treasury interest rate, $\Delta Slope_t$ the change in the slope of the term structure, ΔVIX_t the change in VIX_t index, RM_t the S&P 500 return, and $\Delta Jump_t$ the change in a jump factor based on S&P 500 index options. Columns (1)–(6) report bonds sorted into the six leverage cohorts used throughout the paper, and Column (7) the full sample. The table reports the average coefficients across bonds, the associated t-statistics, the mean and median adjusted R^2 values, and the numbers of observations and bonds in the sample, respectively. The t-statistics are calculated from the cross-sectional variation over the estimates for each coefficient. That is, each reported coefficient value is divided by the standard deviation of the estimates and scaled by the square root of the number of bonds. The sample is based on U.S. industrial corporate bond transaction data from TRACE for the period 2004–2021.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	<15%	15%-25%	25%-35%	35%-45%	45%-55%	>55%	All
Intercept	0.009 (3.462)	0.008 (3.237)	0.018 (5.971)	0.014 (3.609)	0.041 (5.650)	0.065 (7.846)	0.022 (12.281)
$\Delta Lev_{i,t}$	0.003 (0.536)	0.013 (9.510)	0.016 (8.389)	0.038 (15.089)	0.052 (15.668)	0.114 (17.551)	0.033 (21.010)
ΔRF_t	-0.277 (-22.315)	-0.386 (-33.083)	-0.496 (-28.817)	-0.719 (-22.610)	-0.819 (-13.496)	-1.094 (-16.342)	-0.564 (-43.563)
ΔRF_t^2	0.119 (3.007)	0.172 (4.520)	0.087 (1.935)	0.193 (3.038)	0.127 (1.320)	0.209 (1.784)	0.151 (5.953)
$\Delta Slope_t$	0.269 (13.540)	0.364 (23.686)	0.447 (17.822)	0.633 (13.339)	0.825 (9.212)	0.870 (9.449)	0.504 (28.210)
ΔVIX_t	0.003 (1.391)	0.005 (7.144)	0.008 (5.325)	0.009 (5.402)	0.009 (2.861)	0.008 (2.531)	0.006 (8.508)
RM_t	-0.016 (-10.059)	-0.019 (-16.262)	-0.030 (-20.123)	-0.041 (-18.917)	-0.062 (-16.225)	-0.089 (-20.071)	-0.037 (-38.023)
$\Delta Jump_t$	0.005 (5.194)	0.008 (13.169)	0.011 (12.106)	0.018 (12.333)	0.021 (7.398)	0.031 (10.954)	0.014 (23.233)
Mean Adj. R^2	0.306	0.329	0.348	0.408	0.425	0.392	0.355
Median Adj. R^2	0.323	0.352	0.358	0.436	0.450	0.408	0.378
Obs.	81215	124240	97662	57243	32,611	56817	449788
Bonds	1261	1900	1288	878	515	984	6826

Table IA.III: **Principal Component Analysis**

For each industrial bond i with at least 25 monthly observations of yield spread changes, $\Delta YS_{i,t}$, I estimate the model:

$$\Delta YS_{i,t} = \alpha_i + \boldsymbol{\beta}_i^T \boldsymbol{\Delta S}_{i,t} + \varepsilon_{i,t},$$

where $\boldsymbol{\Delta S}_{i,t} := [\Delta Lev_{i,t}, \Delta RF_t, \Delta RF_t^2, \Delta Slope_t, \Delta VIX_t, RM_t, \Delta Jump_t]$ is the vector of the structural model variables, with $\Delta Lev_{i,t}$ as the change in firm leverage, ΔRF_t the change in 10-year Treasury interest rate, ΔRF_t^2 the squared change in the 10-year Treasury interest rate, $\Delta Slope_t$ the change in the slope of the term structure, ΔVIX_t the change in VIX_t index, RM_t the S&P 500 return, and $\Delta Jump_t$ the change in a jump factor based on S&P 500 index options. I then assign each month's residuals to one of 18 bins defined by three maturity groups (less than five years, five to eight years, and over eight years) and six leverage groups (less than 15%, 15%–25%, 25%–35%, 35%–45%, 45%–55%, and greater than 55%) and compute an average residual. I extract the principal components of the covariance matrix of these residuals. For each bin, I report the number of bonds, the number of observations, the principal components loadings, and the ratio of variation of the residual to the total variation. I further report the proportions of the variance of the residuals explained by the first and second principal components, PC1 and PC2, respectively, and the total unexplained variance of the regression in squared percentage points. The sample is based on U.S. industrial corporate bond transaction data from TRACE for the period 2004–2021.

Leverage	Maturity	Bonds	Observations	PC1	PC2	Exp
1	1	913	36914	0.094	0.065	0.011
1	2	642	16399	0.096	0.114	0.011
1	3	702	27902	0.070	0.094	0.006
2	1	1331	51716	0.130	0.140	0.019
2	2	1052	25877	0.136	0.141	0.018
2	3	1151	46647	0.094	0.132	0.010
3	1	877	33402	0.185	0.084	0.035
3	2	785	20118	0.156	0.161	0.024
3	3	813	44142	0.129	0.124	0.017
4	1	622	21038	0.268	0.019	0.070
4	2	597	15212	0.234	0.261	0.053
4	3	539	20993	0.175	0.159	0.033
5	1	403	13519	0.329	-0.585	0.117
5	2	372	9503	0.308	0.162	0.090
5	3	262	9589	0.190	0.100	0.039
6	1	796	25848	0.440	-0.330	0.180
6	2	721	17301	0.401	-0.267	0.153
6	3	398	13668	0.326	0.465	0.115
Proportion of Variance				0.797	0.054	
Unexplained Variance					1.124	

Table IA.IV: **Heterogeneity by Firm Fundamentals**

For each industrial bond i with at least 25 monthly observations of yield spread changes $\Delta YS_{i,t}$, I estimate one bond-fixed-effects panel regression per heterogeneity dimension,

$$\Delta YS_{i,t} = \alpha_i + \beta^T \Delta S_{i,t} + \Gamma^T \Delta C_{i,t} + \sum_g (\theta_g^T \Delta I_{i,t}) \mathbb{1}\{G_{i,t} = g\} + v_{i,t},$$

where $\Delta S_{i,t}$ contains the [CDGM](#) structural credit-risk variables, $\Delta C_{i,t}$ the intermediation controls from [FN](#), [HKS](#), and [EHL](#), and $\Delta I_{i,t}$ the inflation-risk proxies ($\Delta E[\mu^S]_{i,t}$, $\Delta \sigma_{i,t}^S$, $\Delta \rho_t^S$) introduced in Section [III](#). The control coefficients are common across groups, while the inflation-risk coefficients vary with the sorting characteristic. Panels A–D report results for issue size, profitability, profit margin, and credit rating, respectively. Size is measured at issuance; the remaining characteristics are measured one month before the observation, with Compustat reporting timing handled as described in Section [IV](#). The table reports the group-specific coefficients and the High-Minus-Low difference between the extreme groups. t -statistics are reported in parentheses using standard errors two-way clustered by bond and month. Observations and bonds are reported once per dimension, on the High-Minus-Low row.

Panel A: Size					
Group	$\Delta E[\mu^S]_{i,t}$	$\Delta \sigma_{i,t}^S$	$\Delta \rho_t^S$	Obs.	Bonds
Small	-0.451*** (-5.48)	0.473** (1.98)	0.302** (1.99)		
2	-0.350*** (-4.23)	0.764*** (3.33)	0.210 (1.54)		
3	-0.402*** (-3.89)	0.785** (2.02)	0.324** (2.04)		
4	-0.231*** (-3.16)	0.413** (2.13)	0.093 (0.73)		
Large	-0.209** (-2.34)	0.497* (1.89)	-0.068 (-0.33)		
High – Low	0.242*** (3.24)	0.024 (0.08)	-0.370 (-1.50)	449788	6826
Panel B: ROA					
Low	-1.018*** (-7.82)	1.093** (2.30)	0.539** (2.28)		
2	-0.379*** (-4.21)	0.590* (1.89)	0.329* (1.69)		
3	-0.190* (-1.95)	0.605* (1.95)	0.199 (1.46)		
4	-0.013 (-0.24)	0.381** (2.53)	-0.019 (-0.17)		
High	0.005 (0.10)	-0.005 (-0.03)	-0.143 (-1.01)		
High – Low	1.023*** (8.69)	-1.098** (-2.27)	-0.682*** (-2.71)	424180	6826
Panel C: Margin					
Low	-0.961*** (-8.07)	1.098** (2.38)	0.585** (2.51)		
2	-0.322*** (-3.65)	0.468* (1.95)	0.230 (1.60)		
3	-0.139** (-2.42)	0.185 (1.22)	0.131 (0.96)		
4	-0.070 (-0.89)	0.595*** (2.62)	0.108 (0.96)		
High	-0.074 (-1.16)	0.353* (1.68)	-0.144 (-1.23)		
High – Low	0.887*** (8.41)	-0.746* (-1.66)	-0.729*** (-3.18)	423975	6823
Panel D: Rating					
AAA-AA	0.200*** (3.84)	-0.120 (-0.60)	-0.336* (-1.65)		
A	0.081 (1.64)	-0.006 (-0.04)	-0.073 (-0.46)		
BBB	-0.164* (-1.88)	0.652*** (2.79)	0.154 (1.41)		
BB	-0.796*** (-6.42)	1.032* (1.82)	0.580** (2.22)		
B-C	-1.561*** (-8.75)	1.576* (1.95)	0.746 (1.60)		
High – Low	-1.761*** (-9.37)	1.696* (1.82)	1.081* (1.88)	424180	6826

Table IA.V: **Heterogeneity by Operating Channels**

For each industrial bond i with at least 25 monthly observations of yield spread changes $\Delta YS_{i,t}$, I estimate one bond-fixed-effects panel regression per heterogeneity dimension,

$$\Delta YS_{i,t} = \alpha_i + \beta^T \Delta S_{i,t} + \Gamma^T \Delta C_{i,t} + \sum_g (\theta_g^T \Delta I_{i,t}) \mathbb{1}\{G_{i,t} = g\} + v_{i,t},$$

where $\Delta S_{i,t}$ contains the [CDGM](#) structural credit-risk variables, $\Delta C_{i,t}$ the intermediation controls from [FN](#), [HKS](#), and [EHL](#), and $\Delta I_{i,t}$ the inflation-risk proxies ($\Delta E[\mu^S]_{i,t}$, $\Delta \sigma_{i,t}^S$, $\Delta \rho_t^S$) introduced in Section [III](#). The control coefficients are common across groups, while the inflation-risk coefficients vary with the sorting characteristic. Panels A–C report results for operating leverage, markup, and price pass-through, respectively. Operating leverage and markup are measured one month before the observation, while price pass-through uses the expanding mean of the underlying monthly series through twelve months before the observation, as described in Section [IV](#). t -statistics are reported in parentheses using standard errors two-way clustered by bond and month.

Panel A: Op. Leverage					
Group	$\Delta E[\mu^S]_{i,t}$	$\Delta \sigma_{i,t}^S$	$\Delta \rho_t^S$	Obs.	Bonds
Low	-0.347*** (-3.42)	0.354 (1.24)	0.116 (0.82)		
2	-0.374*** (-4.73)	0.551** (2.19)	0.172 (1.01)		
3	-0.325*** (-3.67)	0.429* (1.73)	0.182 (1.48)		
4	-0.325*** (-3.38)	0.894*** (2.73)	0.180 (1.45)		
High	-0.295*** (-3.60)	0.608*** (2.59)	0.147 (1.44)		
High – Low	0.052 (0.55)	0.255 (0.95)	0.031 (0.29)	424180	6826
Panel B: Markup					
Low	-0.550*** (-6.18)	0.568 (1.59)	0.262 (1.55)		
2	-0.315*** (-3.19)	0.452* (1.89)	0.168 (1.09)		
3	-0.294*** (-3.52)	0.536** (2.16)	0.153 (1.11)		
4	-0.246*** (-3.47)	0.712*** (3.06)	0.138 (1.20)		
High	-0.255*** (-3.06)	0.571** (2.42)	0.099 (1.09)		
High – Low	0.296*** (3.99)	0.003 (0.01)	-0.163 (-1.20)	423975	6823
Panel C: Price pass-through					
Low	-0.175** (-2.53)	0.009 (0.05)	0.132 (0.89)		
2	-0.103* (-1.88)	0.362** (2.03)	-0.073 (-0.76)		
3	-0.044 (-0.78)	0.613*** (2.79)	-0.023 (-0.27)		
4	-0.196** (-2.23)	0.466* (1.88)	-0.216 (-1.42)		
High	-0.615*** (-4.81)	0.935* (1.87)	0.377 (1.09)		
High – Low	-0.440*** (-3.47)	0.925* (1.79)	0.245 (0.79)	108686	2749

Table IA.VI: **Heterogeneity by Debt Structure**

For each industrial bond i with at least 25 monthly observations of yield spread changes $\Delta YS_{i,t}$, I estimate one bond-fixed-effects panel regression per heterogeneity dimension,

$$\Delta YS_{i,t} = \alpha_i + \beta^T \Delta S_{i,t} + \Gamma^T \Delta C_{i,t} + \sum_g (\theta_g^T \Delta I_{i,t}) \mathbb{1}\{G_{i,t} = g\} + v_{i,t},$$

where $\Delta S_{i,t}$ contains the **CDGM** structural credit-risk variables, $\Delta C_{i,t}$ the intermediation controls from **FN**, **HKS**, and **EHL**, and $\Delta I_{i,t}$ the inflation-risk proxies ($\Delta E[\mu^S]_{i,t}$, $\Delta \sigma_{i,t}^S$, $\Delta \rho_t^S$) introduced in Section III. The control coefficients are common across groups, while the inflation-risk coefficients vary with the sorting characteristic. Panels A–E report results for leverage, refinancing intensity, debt growth, floating-rate exposure, and maturity, respectively. Leverage, refinancing intensity, floating-rate exposure, and maturity are measured one month before the observation, while debt growth uses the expanding mean of the underlying monthly series through twelve months before the observation, as described in Section IV. t -statistics are reported in parentheses using standard errors two-way clustered by bond and month.

Panel A: Leverage						
Group	$\Delta E[\mu^S]_{i,t}$	$\Delta \sigma^S_{i,t}$	$\Delta \rho^S_t$	Obs.	Bonds	
<15%	0.085** (2.03)	-0.009 (-0.06)	-0.120 (-0.93)			
15%-25%	0.044 (0.90)	0.125 (0.95)	-0.038 (-0.33)			
25%-35%	-0.125** (-2.13)	0.351** (2.13)	0.191* (1.87)			
35%-45%	-0.164** (-2.01)	0.467** (2.56)	-0.101 (-0.61)			
45%-55%	-0.429*** (-4.40)	0.760*** (3.00)	0.055 (0.31)			
>55%	-1.309*** (-7.68)	1.483** (2.11)	1.016** (2.07)			
High – Low	-1.394*** (-8.31)	1.492* (1.94)	1.136** (2.07)	424180	6826	
Panel B: Ref. Intensity						
Low	-0.611*** (-5.05)	1.185** (2.41)	0.413*** (2.80)			
2	-0.386*** (-4.45)	1.005*** (3.01)	0.194 (1.23)			
3	-0.236*** (-3.00)	0.269 (1.21)	0.000 (0.00)			
4	-0.172* (-1.92)	0.327 (1.24)	-0.014 (-0.08)			
High	-0.366*** (-4.19)	0.492** (2.25)	0.186 (1.19)			
High – Low	0.245** (2.49)	-0.694* (-1.66)	-0.227 (-1.54)	420299	6788	
Panel C: Debt Growth						
Low	-0.253*** (-4.24)	0.415* (1.95)	0.259 (1.54)			
2	-0.296*** (-3.28)	0.607** (2.03)	0.177 (1.20)			
3	-0.247*** (-3.40)	0.411** (2.14)	-0.036 (-0.32)			
4	-0.333*** (-4.16)	0.655** (2.01)	0.117 (0.82)			
High	-0.504*** (-4.46)	0.584* (1.92)	-0.001 (0.00)			
High – Low	-0.251** (-2.35)	0.169 (0.49)	-0.260 (-1.62)	293639	6824	
Panel D: Float						
Fixed	-0.357*** (-4.70)	0.605*** (2.72)	0.152 (1.23)			
Float	-0.154* (-1.86)	0.325* (1.69)	0.233* (1.96)			
High – Low	0.203*** (4.00)	-0.280* (-1.76)	0.082 (0.90)	424180	6826	
Panel E: Maturity						
<5	-0.314*** (-4.76)	0.413** (2.26)	0.540*** (2.93)			
5-8	-0.528*** (-3.65)	1.216*** (2.89)	0.031 (0.20)			
>8	-0.287** (-2.32)	0.688 (1.62)	-0.196 (-1.28)			
High – Low	0.027 (0.28)	0.275 (0.66)	-0.736*** (-2.81)	424180	6826	

Table IA.VII: **Heterogeneity by Trading Volume**

For each industrial bond i with at least 25 monthly observations of yield spread changes $\Delta YS_{i,t}$, I estimate a bond-fixed-effects panel regression,

$$\Delta YS_{i,t} = \alpha_i + \beta^T \Delta S_{i,t} + \Gamma^T \Delta C_{i,t} + \sum_g (\theta_g^T \Delta I_{i,t}) \mathbb{1}\{G_{i,t} = g\} + v_{i,t},$$

where $\Delta S_{i,t}$ contains the [CDGM](#) structural credit-risk variables, $\Delta C_{i,t}$ the intermediation controls from [FN](#), [HKS](#), and [EHL](#), and $\Delta I_{i,t}$ the inflation-risk proxies ($\Delta E[\mu^S]_{i,t}$, $\Delta \sigma_{i,t}^S$, $\Delta \rho_t^S$) introduced in Section [III](#). The control coefficients are common across groups, while the inflation-risk coefficients vary across quintiles of monthly trading volume measured one month before the observation. The intermediation controls already include bond- and market-level liquidity and friction measures. t -statistics are reported in parentheses using standard errors two-way clustered by bond and month. Observations and bonds are reported once, on the High-Minus-Low row.

Group	$\Delta E[\mu^S]_{i,t}$	$\Delta \sigma_{i,t}^S$	$\Delta \rho_t^S$	Obs.	Bonds
Low	-0.338*** (-4.01)	0.272 (1.01)	0.145 (1.54)		
2	-0.186*** (-2.78)	0.753*** (3.76)	0.167 (1.63)		
3	-0.284*** (-3.42)	0.566** (2.51)	0.193* (1.65)		
4	-0.389*** (-4.26)	0.708*** (2.96)	0.185 (1.28)		
High	-0.504*** (-5.84)	0.595** (2.24)	0.110 (0.59)		
High – Low	-0.165** (-2.47)	0.323 (1.37)	-0.035 (-0.22)	424180	6826

Table IA.VIII: **Price Pass-Through: Measurement and Cross-Sectional Evidence**

This table examines price pass-through and heterogeneity in expected-inflation sensitivity. Panel A relates the price-pass-through measure to realized margins. Price pass-through is the absolute ratio of an industry's output-price inflation to the input-weighted producer-price inflation of its supplier industries. I regress quarterly changes in industry gross margins on inflation interacted with pass-through terciles, including industry and quarter fixed effects. Gross margins are constructed from sales and cost of goods sold aggregated across Compustat firms within four-digit NAICS industries. Panels B and C examine bond-level heterogeneity. For each industrial bond i with at least 25 monthly observations of yield spread changes $\Delta YS_{i,t}$, I estimate bond-fixed-effects panel regressions including the [CDGM](#) structural credit-risk controls, the intermediation controls from [FN](#), [HKS](#), and [EHL](#), and the inflation-risk proxies introduced in Section [III](#). Panel B double-sorts bonds into a 3×3 grid based on markup and price pass-through and reports High-Minus-Low differences holding the other sort fixed. Panel C repeats the pass-through comparison among low-markup firms within investment-grade and high-yield bonds and, separately, in the full pass-through sample within manufacturing and non-manufacturing firms. t -statistics are reported in parentheses using standard errors clustered by industry and quarter in Panel A and two-way clustered by bond and month in Panels B–C.

Panel A: Price Pass-Through and Industry Gross Margins				
$\partial \Delta \text{Gross margin} / \partial \text{inflation}$	Low	Mid	High	High – Low
<i>Manufacturing only</i>				
Own-ind output PPI inflation	0.007 (0.31)	0.003 (0.05)	0.065* (1.90)	0.058 (0.90)
Input-cost inflation	0.029 (1.47)	-0.004 (-0.09)	0.057 (1.28)	0.027 (0.45)
<i>All industries</i>				
Own-ind output PPI inflation	0.002 (0.05)	-0.002 (-0.08)	0.083*** (3.07)	0.081 (1.19)
Input-cost inflation	0.005 (0.14)	-0.011 (-0.32)	0.063** (2.34)	0.058 (1.02)
Panel B: Expected-Inflation Sensitivity by Markup and Price Pass-Through				
	Low pass-through	Mid pt	High pt	High – Low
Low markup	-0.323*** (-3.00)	-0.095 (-1.27)	-0.736*** (-5.28)	-0.413*** (-3.54)
Mid markup	-0.087 (-1.10)	0.076 (1.23)	-0.004 (-0.04)	0.084 (0.59)
High markup	0.042 (0.72)	-0.026 (-0.45)	0.000 (-0.01)	-0.043 (-0.63)
Markup High – Low	0.366*** (3.14)	0.068 (0.84)	0.736*** (5.91)	
Panel C: Price Pass-Through across Credit Quality and Industry				
Sample	$\Delta E[\mu^S]_{i,t}$	Obs.	Bonds	
<i>within low markup, by rating</i>				
Investment grade	-0.338*** (-3.10)	33274	1027	
High yield	-0.342 (-1.39)	14932	523	
<i>full pass-through sample, by industry</i>				
All industries	-0.309*** (-3.06)	105007	2739	
Manufacturing only	-0.302*** (-2.76)	96094	2523	
Non-manufacturing	-0.355 (-1.44)	8913	347	

Table IA.IX: Inflation-Risk Sensitivity and Importance by Subperiod

For each industrial bond i with at least 25 monthly observations of yield spread changes $\Delta YS_{i,t}$, I estimate the fully controlled bond-fixed-effects specification including the [CDGM](#) structural credit-risk variables, the intermediation controls from [FN](#), [HKS](#), and [EHL](#), and the inflation-risk proxies ($\Delta E[\mu^S]_{i,t}$, $\Delta \sigma_{i,t}^S$, $\Delta \rho_i^S$) introduced in Section III. I re-estimate the specification separately over three subperiods: September 2004–December 2007, January 2008–December 2019, and January 2020–December 2021. Panel A reports the coefficients on the three inflation-risk proxies, with t -statistics in parentheses based on standard errors two-way clustered by bond and month. Panel B reports the explanatory power of the inflation-risk proxies, showing the within- R^2 of the control specification, the within- R^2 after adding all three proxies, and the resulting increase. The first and last subperiods each contain roughly one-eighth of the full sample and therefore yield less precise estimates. The sample is based on U.S. industrial corporate bond transaction data from TRACE for the period 2004–2021.

	Full sample	Pre-GFC	Post-GFC	Post-COVID
<i>Panel A: coefficient on each proxy</i>				
$\Delta E[\mu^S]_{i,t}$	-0.339*** (-4.53)	-0.069 (-1.18)	-0.311*** (-3.68)	-0.500*** (-3.29)
$\Delta \sigma_{i,t}^S$	0.587*** (2.86)	-0.127 (-0.40)	0.462** (2.01)	1.498*** (3.78)
$\Delta \rho_i^S$	0.167 (1.40)	-0.010 (-0.11)	0.344** (2.56)	-0.910 (-1.49)
<i>Panel B: within-R^2, controls versus controls plus proxies</i>				
Controls only	0.204	0.139	0.163	0.367
Adding the three proxies	0.215	0.139	0.174	0.382
gain	0.011	0.000	0.011	0.015
Obs.	449788	51931	337473	60384
Bonds	6826	2052	6619	3278

Table IA.X: Inflation Risk and Yield Spread Changes: All Bonds

For each bond i with at least 25 monthly observations of yield spread changes $\Delta YS_{i,t}$, I estimate the model

$$\Delta YS_{i,t} = \alpha_i + \beta_i^T \Delta S_{i,t} + \theta_i^T \Delta I_{i,t} + \Gamma_i^T \Delta C_{i,t} + v_{i,t},$$

where $\Delta S_{i,t}$ represents structural model variables, $\Delta I_{i,t}$ denotes inflation-risk proxies, and $\Delta C_{i,t}$ refers to control proxies from [Friewald and Nagler \(2019\)](#), [He, Khorrami and Song \(2022\)](#) and [Eisfeldt, Herskovic and Liu \(2024\)](#). Panel A presents average coefficients, t-statistics, mean and median adjusted R^2 values, and sample sizes. Panel B details a principal component analysis on the residuals, reporting the variance explained by the first two principal components and total unexplained variance. Panel C includes R^2 values, F-statistics, and p-values from a Wald-test of the time-series regression of PC1 on inflation-risk proxies. The sample is U.S. corporate bond transaction data from TRACE for the period 2004–2021, with no restriction on industry, issue size, or bond type.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Individual Bond Regressions								
$\Delta E[\mu^S]_{i,t}$		-0.457 (-46.939)			-0.418 (-45.475)	-0.309 (-29.990)	-0.317 (-29.224)	-0.250 (-22.892)
$\Delta \sigma_{i,t}^S$			1.400 (45.182)		1.120 (37.637)	0.921 (27.137)	0.938 (24.632)	0.586 (14.193)
$\Delta \rho_i^S$				0.377 (22.843)	0.097 (6.522)	0.028 (1.450)	0.012 (0.587)	0.022 (1.025)
CDGM	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
FN	No	No	No	No	No	Yes	Yes	Yes
HKS	No	No	No	No	No	No	Yes	Yes
EHL	No	No	No	No	No	No	No	Yes
Mean Adj. R^2	0.339	0.370	0.401	0.374	0.431	0.501	0.513	0.525
Median Adj. R^2	0.358	0.390	0.406	0.385	0.439	0.531	0.549	0.568
Obs.	911634	911634	911634	911634	911634	911634	911634	911634
Bonds	14338	14338	14338	14338	14338	14338	14338	14338
Panel B: Principal Component Analysis								
FVE		0.146	0.197	0.098	0.336	0.259	0.271	0.257
PC1	0.765	0.737	0.732	0.763	0.716	0.723	0.717	0.683
PC2	0.135	0.143	0.138	0.129	0.142	0.112	0.116	0.131
UV	0.658	0.562	0.528	0.593	0.437	0.223	0.202	0.156
Panel C: Time-Series Regression of PC1 on Added Variables								
Adj. R^2		0.083	0.134	0.044	0.160	0.088	0.078	0.040
R^2		0.088	0.138	0.048	0.172	0.101	0.091	0.054
F-stat		19.816	32.898	10.415	14.146	7.678	6.847	3.873
p-value		0.000	0.000	0.001	0.000	0.000	0.000	0.010
Obs.		208	208	208	208	208	208	208

Table IA.XI: **Inflation Risk and Yield Spread Changes: End of Month Only**

For each industrial bond i with at least 25 monthly observations of yield spread changes $\Delta YS_{i,t}$, I estimate the model

$$\Delta YS_{i,t} = \alpha_i + \beta_i^T \Delta S_{i,t} + \theta_i^T \Delta I_{i,t} + \Gamma_i^T \Delta C_{i,t} + v_{i,t},$$

where $\Delta S_{i,t}$ represents structural model variables, $\Delta I_{i,t}$ denotes inflation-risk proxies, and $\Delta C_{i,t}$ refers to control proxies from [Friewald and Nagler \(2019\)](#), [He, Khorrami and Song \(2022\)](#) and [Eisfeldt, Herskovic and Liu \(2024\)](#). Panel A presents average coefficients, t-statistics, mean and median adjusted R^2 values, and sample sizes. Panel B details a principal component analysis on the residuals, reporting the variance explained by the first two principal components and total unexplained variance. Panel C includes R^2 values, F-statistics, and p-values from a Wald-test of the time-series regression of PC1 on inflation-risk proxies. The sample is based on U.S. industrial corporate bond transaction data from TRACE for the period 2004–2021.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Individual Bond Regressions								
$\Delta E[\mu^S]_{i,t}$		-0.733 (-15.087)			-0.617 (-16.161)	-0.506 (-9.301)	-0.519 (-9.522)	-0.448 (-8.888)
$\Delta \sigma_{i,t}^S$			2.018 (13.349)		1.517 (12.078)	1.187 (8.472)	1.100 (7.607)	0.360 (2.558)
$\Delta \rho_t^S$				0.747 (9.200)	0.268 (4.077)	0.005 (0.057)	-0.071 (-0.830)	-0.110 (-1.330)
CDGM	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
FN	No	No	No	No	No	Yes	Yes	Yes
HKS	No	No	No	No	No	No	Yes	Yes
EHL	No	No	No	No	No	No	No	Yes
Mean Adj. R^2	0.386	0.418	0.454	0.427	0.482	0.545	0.556	0.568
Median Adj. R^2	0.414	0.449	0.473	0.455	0.505	0.585	0.601	0.613
Obs.	236694	236694	236694	236694	236694	236694	236694	236694
Bonds	4356	4356	4356	4356	4356	4356	4356	4356
Panel B: Principal Component Analysis								
FVE		0.136	0.242	0.100	0.355	0.352	0.365	0.306
PC1	0.611	0.546	0.574	0.585	0.502	0.494	0.516	0.503
PC2	0.127	0.151	0.127	0.124	0.160	0.158	0.156	0.170
UV	4.005	3.460	3.035	3.604	2.582	1.487	1.234	1.135
Panel C: Time-Series Regression of PC1 on Added Variables								
Adj. R^2		0.065	0.051	0.033	0.090	0.048	0.036	0.029
R^2		0.069	0.055	0.038	0.103	0.062	0.050	0.043
F-stat		15.385	12.019	8.099	7.799	4.471	3.568	3.057
p-value		0.000	0.001	0.005	0.000	0.005	0.015	0.029
Obs.		208	208	208	208	208	208	208